

Credible on Paper? Evaluating the Effect of Green Banking Policy on Firms' Green Investment in Emerging Economies

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Abstract

More than sixty emerging and developing economies have adopted green banking guidelines through the Sustainable Banking and Finance Network (SBFN). The policy's theory of change holds that supervisory guidelines lead banks to price environmental risk, so that credit flows to firms investing in cleaner production. We evaluate two links in that chain. At firm level, using two independent World Bank Enterprise Survey samples (28,042 firms in 41 economies; 89,797 firms in 162 economies) and four estimators, bank credit is associated with 13-percentage-point higher green-practice adoption, but the association is the same in adopting and non-adopting economies (0.076 versus 0.072 in a causal forest) and does not vary with regulatory quality. At country level, a staggered-adoption event study of 217 economies finds no effect on renewable-energy share, and a CO₂ association that appears to reflect which economies join. The guidelines show no detectable effect on the margin they target.

Keywords: green banking policy, policy evaluation, theory of change, access to finance, staggered difference-in-differences, causal forest

JEL: G21, O16, Q56, D22, C11

1. Introduction

Since 2012, more than sixty central banks and financial regulators across the emerging and transition world have signed on to a coordinated effort to green their financial systems through the IFC’s Sustainable Banking and Finance Network (SBFN), directing commercial banks to price environmental risk into lending and channel credit toward cleaner production. The network monitors this effort by tracking policy outputs: which guidelines each member has issued, and how far along its own progression framework each jurisdiction has moved. It does not track whether firms change what they do. Whether adoption changes what happens on the ground—whether a firm that can get a bank loan is any more likely to install pollution-control equipment or measure its own emissions—is the question an evaluation of the policy has to answer, and it is the one this paper takes up.

We approach SBFN adoption as a policy intervention and evaluate it against its own theory of change (Section 2.1). That theory runs from a supervisory guideline, through banks pricing environmental risk and reallocating credit, to financed firms investing more in cleaner production, and from there to better aggregate environmental outcomes. If the chain works, two things should be observable: the link between bank credit and firms’ green investment should be stronger where the guideline is in force, and adopting economies’ environmental trajectories should improve after adoption. We test both, with enough statistical power to distinguish a true absence of effect from an underpowered test of one.

A large literature on state capacity argues that a formally identical policy can produce different real-economy outcomes depending on the administra-

tive infrastructure behind it (La Porta et al., 1998; Acemoglu et al., 2001),
 and recent country-level evidence finds green-finance policy effectiveness itself
 conditional on governance quality (Jia et al., 2026; Chi and Yang, 2023; Shi
 et al., 2024; W. Ullah et al., 2024). Each of these results implies a firm-level
 prediction—that the credit-to-green channel should respond to policy more
 sharply where regulatory capacity is stronger—that has not been tested di-
 rectly, across a broad cross-section of adopting economies, on firm-level data.
 A separate strand shows firm-level financing constraints shape green invest-
 ment (Ghosh and Dutta, 2022; Costa et al., 2024; De Haas et al., 2025) and
 that the WBES Green Economy module is a workable lens on firm environ-
 mental behavior (Asif et al., 2025; Phan, 2026), but nothing yet connects the
 two literatures at firm level.

We test this using two independently assembled World Bank Enterprise
 Survey (WBES) samples: a 41-economy primary sample (28,042 firms, 2018–
 2023) carrying the Bank’s full Green Economy module, and a 162-economy
 global sample (89,797–100,115 firms, 2021–2026) reaching, for the first time
 in this literature, into high-income economies entirely outside SBFN’s remit
 (Section 4). We match each economy to its SBFN policy status as of its sur-
 vey year and to the World Bank’s Worldwide Governance Indicators. Because
 SBFN status is fixed within country in a single wave, a country fixed effect
 absorbs its main effect entirely; its interaction with credit access remains
 formally identified, but off only the eight adopter economies in the primary
 sample, which is why we also estimate a Bayesian hierarchical model with
 country-varying slopes (Gelman and Hill, 2007; Gelman et al., 2020) and a
 causal-forest / double-machine-learning estimator (Wager and Athey, 2018;

51 Chernozhukov et al., 2018; Athey et al., 2019; Athey and Imbens, 2019) that
52 let the data reveal moderation rather than relying on a single fragile inter-
53 action term. On the larger global sample, country fixed effects are usable
54 directly. Across all four estimators the answer is the same, and precisely so:
55 the causal forest’s credit effect differs by only 0.004 between SBFN adopters
56 and non-adopters, and is flat across regulatory-quality terciles—these are
57 tight estimates that rule out policy-relevant moderation, not underpowered
58 nulls that merely fail to detect one.

59 This paper contributes to the evaluation of green finance policy in four
60 respects. First, it is, to our knowledge, the first evaluation of SBFN-style
61 adoption against firm-level outcomes across a broad cross-section of adopt-
62 ing economies, rather than against the policy-stage indicators the network
63 itself reports or the bank-level lending data that dominate the green-credit
64 literature. Second, it tests the policy’s theory of change at two points, the
65 firm-level credit channel and the aggregate environmental outcome, using dif-
66 ferent data, units of analysis and identification strategies; the country-level
67 test (Section 6) is a Callaway-Sant’Anna staggered-adoption event study ex-
68 ploiting SBFN’s own join dates. Third, the firm-level result is replicated on
69 an independently assembled 162-economy global sample with almost no over-
70 lap with the primary sample, a different outcome measure and country fixed
71 effects. Fourth, because the estimates are precise rather than merely insignif-
72 icant, the evaluation can report a genuine absence of effect on the targeted
73 margin, the kind of null finding that is informative for programme design but
74 rarely published. The heterogeneity that does exist sits at firm size rather
75 than country institutions (Section 3.4); we report it as exploratory, since it

76 was not pre-specified.

77 Section 2 describes the SBFN framework and sets out its theory of change;
78 Section 3 develops the hypotheses; Section 4 describes data and empirical
79 strategy; Section 5 evaluates the firm-level credit channel; Section 6 evaluates
80 aggregate environmental outcomes; Section 7 draws lessons for policy and for
81 evaluation; and Section 8 concludes.

82 **2. Institutional background: green banking policy in emerging and** 83 **transition economies**

84 The Sustainable Banking and Finance Network (SBFN)—established in
85 2012 as the Sustainable Banking Network and renamed in 2019 to reflect an
86 expanded mandate covering both banking and non-bank financial regulation—
87 is the principal vehicle through which financial-sector regulators and banking
88 associations in emerging and developing economies have coordinated the de-
89 sign of national sustainable-finance policy. Hosted and technically supported
90 by the International Finance Corporation, the network now counts roughly
91 seventy member institutions (central banks, banking regulators, and banking
92 or securities associations) drawn overwhelmingly from Sub-Saharan Africa,
93 East Asia and the Pacific, Europe and Central Asia, Latin America and
94 the Caribbean, the Middle East and North Africa, and South Asia (Inter-
95 national Finance Corporation / Sustainable Banking and Finance Network,
96 2025). Notably for our sample, the network’s membership is essentially dis-
97 joint from the European Union: not one of the sixteen EU member states in
98 our WBES sample appears on any SBFN membership roster we could locate,
99 consistent with the network’s explicit mandate to serve markets outside the

100 reach of the EU’s own sustainable-finance architecture (the Sustainable Fi-
101 nance Disclosure Regulation and the EU Taxonomy, both of which post-date
102 most of our survey waves in any case).

103 Membership itself, however, is only the network’s coarsest measure of
104 policy adoption. SBFN’s own Multi-Tiered Progress Framework—the in-
105 strument used in its periodic Global Progress Reports (2018, 2019, 2021,
106 2023–2024) and country-level Progress Report addenda—scores each mem-
107 ber jurisdiction’s actual regulatory output against several policy pillars, most
108 centrally environmental-and-social risk management and, in later report vin-
109 tages, ESG disclosure integration and green-finance-flow measurement, and
110 places each jurisdiction on a progression from an initial “Commitment” or
111 “Formulating” stage, through “Developing” or “Implementation,” to “Advanc-
112 ing” and, for the most mature frameworks, “Consolidating.” The variation
113 this produces across our sample is considerable and is itself part of the paper’s
114 identifying variation, not just a robustness afterthought. Mongolia’s Bank
115 of Mongolia was a founding SBFN member in 2012 and had already reached
116 the network’s “Advancing” implementation stage by the time of the country’s
117 2019 Enterprise Survey; Morocco’s Bank Al-Maghrib, a member since 2014,
118 sat at the same stage two years later. Bangladesh Bank, also a founding 2012
119 member, had progressed by its 2022 Enterprise Survey wave into the net-
120 work’s ESG-integration pillar at an “Advancing” classification, while India’s
121 Reserve Bank/Indian Banks’ Association, a later (2016) entrant, remained
122 at a “Developing” commitment-pillar stage at its own 2022 survey. Jordan
123 and the Kyrgyz Republic, both members since 2016 and 2018 respectively,
124 had progressed only as far as an initial “Commitment” stage by their 2019

125 survey waves. A further, and for identification purposes equally important,
126 group of economies in our sample joined SBFN only after their Enterprise
127 Survey was fielded—Kazakhstan (2021), Serbia and Ukraine (2020), Tajik-
128 istan (2023), and most of the Western Balkan and South Caucasus economies
129 in our sample (2021–2024)—and are accordingly coded as non-adopters as of
130 their survey year, even though several are now members. This ex-ante cod-
131 ing is deliberate: it is the policy environment a firm actually operated under
132 at the moment its Green Economy module responses were recorded that is
133 relevant to our test, not the policy environment that jurisdiction eventually
134 adopted.

135 *2.1. The policy’s theory of change and what this evaluation tests*

136 The case that SBFN adoption should change firm behavior, rather than
137 remain a symbolic commitment, rests on a supervisory mechanism that can
138 be written as a chain of five links:

- 139 1. *Adoption.* A banking regulator or banking association issues environmental-
140 and-social risk management guidelines.
- 141 2. *Bank response.* Supervised banks face a compliance incentive to price
142 environmental risk into individual lending decisions and, in the more
143 advanced frameworks, to report a green-lending share to the supervisor.
- 144 3. *Credit allocation.* Credit flows more readily to firms undertaking cleaner
145 investment and less readily to high-polluting borrowers.
- 146 4. *Firm response.* Firms with bank credit invest more in green practices
147 than they would without the guideline, so the link between credit and
148 green investment is stronger where the guideline is in force (H2, Sec-

tion 3), and stronger still where the supervisor has the capacity to enforce it (H3).

5. *Aggregate outcome.* Adopting economies’ environmental indicators improve relative to non-adopters after adoption.

Existing evidence on China’s green credit guidelines documents links 2 to 4 operating at bank and firm level—reduced lending to high-polluting borrowers and improved environmental performance among firms that retain access to credit (Li et al., 2023; Huang et al., 2023; Dai et al., 2025; Gao et al., 2022)—and a smaller literature on Bangladesh’s long-running green-banking guidelines points the same way (Khairunnessa et al., 2021). Neither literature has established whether the chain holds across a broad, heterogeneous cross-section of SBFN-style adopters, or whether its strength depends on the regulatory capacity behind the guideline. This paper tests link 4 at firm level (Section 5) and link 5 at country level (Section 6). It does not observe links 2 and 3 directly: the Enterprise Surveys record whether a firm holds bank credit, not the terms on which banks offered it or what share of bank lending was green. A null at link 4 is therefore informative about whether the chain delivers its intended firm-level effect, but cannot on its own locate the upstream link at which it breaks.

3. Literature review and hypothesis development

3.1. Access to finance and firm-level green investment

A firm’s decision to install pollution-abatement equipment, retrofit machinery for energy efficiency, or absorb the upfront cost of an emissions-monitoring system is, first and foremost, an investment decision, and in-

173 vestment decisions are shaped by whether a firm can finance them. The
 174 theoretical case is close to a standard credit-constraint story: green invest-
 175 ments of the kind captured in the WBES Green Economy module (heat-
 176 ing and cooling upgrades, on-site renewable generation, waste-minimization
 177 retrofits, energy-management systems) typically carry a large upfront capi-
 178 tal outlay against a payback stream realized only over several years, which is
 179 exactly the profile of investment most sensitive to a firm’s access to external
 180 finance rather than to retained earnings alone. Firm-level evidence bears
 181 this out directly. Ghosh and Dutta (2022), using a panel of Indian manu-
 182 facturing firms, show that credit-constrained firms are measurably less likely
 183 to adopt cleaner production practices even when they report an equivalent
 184 willingness to comply with environmental regulation. De Haas et al. (2025)
 185 document, across a large cross-country firm sample, that managerial and fi-
 186 nancial barriers operate as distinct but jointly binding constraints on firms’
 187 green transition investments, and Kalantzis et al. (2022) reach a closely re-
 188 lated conclusion using EBRD transition-economy survey data, finding that
 189 firms’ green investment is driven by a mix of financing conditions and cli-
 190 mate exposure rather than by climate motivation alone—precisely the joint
 191 finance-and-institutions framing this paper tests directly. Costa et al. (2024)
 192 and Truong (2026) find, respectively in OECD and Vietnamese data, that
 193 financing constraints blunt firms’ green investment response to environmen-
 194 tal policy pressure that would otherwise induce it, while Aristei and Gallo
 195 (2023) show that firms with better access to credit are both more likely to
 196 have adopted green management practices before a shock and better able to
 197 sustain them through one. B. Ullah (2025) narrows this further to unlisted

198 SMEs specifically, showing that financial constraints bind corporate environ-
199 mental responsibility at smaller firm scales—a firm-size channel this paper’s
200 own causal-forest evidence (Section 5) also implicates, though, as we report
201 there, without recovering a clean monotonic size gradient of its own. Siewers
202 et al. (2024) extend the firm-level environmental-performance literature to
203 a global-value-chain setting, showing that a firm’s position in international
204 production networks shapes its environmental performance net of domes-
205 tic financing conditions, and Bambe et al. (2026) show that environmental
206 policy stringency itself reshapes firm efficiency in developing economies, un-
207 derscoring that the firm-level margin, not only the aggregate or bank-level
208 one, is where much of this literature’s true test now lies. Taken together, this
209 literature converges on a simple first-order proposition, which any test of a
210 green-credit-policy channel must first establish before asking whether policy
211 moderates it:

212 **H1.** Firms with access to formal bank credit (a credit line or
213 loan from a financial institution) are more likely to have adopted
214 green production practices than otherwise-similar firms without
215 such access.

216 *3.2. Does green banking policy strengthen the finance-to-green-investment*
217 *channel?*

218 H1 describes a channel, not a policy. The entire premise of the SBFN
219 framework and its national analogues—and of the much larger literature on
220 green credit guidelines in China specifically (Li et al., 2023, 2024; Huang et
221 al., 2023)—is that a supervisory mandate directing banks to price environ-
222 mental risk into lending, or to preferentially finance green activity, should

223 widen the gap in green-investment behavior between firms with and without
 224 bank access relative to a counterfactual world without the policy: banks in
 225 adopting jurisdictions have a compliance incentive to seek out, and price
 226 favorably, exactly the borrowers undertaking the kind of investment our
 227 outcome variable measures, while credit-constrained firms in the same ju-
 228 risdiction see no comparable improvement in their financing terms. This
 229 is a moderation hypothesis, not a simple main-effect one, and it is worth
 230 being explicit that most of the existing supporting evidence—again, over-
 231 whelmingly from China (Dai et al., 2025; Gao et al., 2022) plus a smaller
 232 literature on Bangladesh (Khairunnessa et al., 2021), Vietnam (Nguyen and
 233 Phan, 2025), and cross-country fintech-credit channels (Tran et al., 2026)—
 234 is drawn from single-country studies of a single, unusually well-resourced
 235 green-credit regime. Whether the same interaction holds across the far more
 236 heterogeneous set of jurisdictions that have adopted an SBFN-style frame-
 237 work, many of them at a far earlier and less resourced stage of implementation
 238 than China’s, is untested. This motivates:

239 **H2.** The positive association between bank credit access and
 240 green-practice adoption (H1) is stronger in economies that had,
 241 as of the survey year, adopted an SBFN sustainable-finance policy
 242 framework than in economies that had not.

243 *3.3. Institutional complementarity: does policy credibility depend on state*
 244 *capacity?*

245 A green banking guideline is only as effective as the supervisory apparatus
 246 behind it. This is the institutional-complementarity argument at the center of

247 the comparative-institutions, law-and-finance, and state-capacity literatures.
 248 Hall and Soskice (2001)’s varieties-of-capitalism framework is the founda-
 249 tional statement of the underlying logic in its most general form: formally
 250 similar policy instruments produce systematically different outcomes depend-
 251 ing on how they interlock with the surrounding institutional architecture—
 252 a regulatory mandate that functions as intended within one configuration
 253 of complementary institutions may function quite differently, or not at all,
 254 transplanted into another. Applied more narrowly to financial regulation
 255 specifically, the same logic runs through the law-and-finance and state-capacity
 256 literatures: La Porta et al. (1998) and the subsequent finance-and-growth
 257 literature (Levine, 2005; Demirgüç-Kunt and Maksimovic, 1998; Beck et al.,
 258 2005, 2008) established that formally identical financial-contracting rules pro-
 259 duce very different real-economy financing outcomes depending on the quality
 260 of the legal and regulatory institutions that enforce them, while Acemoglu
 261 et al. (2001) showed the same logic operating at the level of a country’s en-
 262 tire institutional endowment. Applied to our setting, the argument is direct:
 263 a bank supervisor’s environmental-risk guideline can only reshape lending
 264 behavior to the extent that the supervisor has the regulatory capacity—
 265 examination authority, credible sanctions, reliable disclosure enforcement—
 266 to make compliance with it more than nominal. Where that capacity is
 267 weak, an SBFN framework may exist on paper (a formal “Commitment” or
 268 “Formulating” stage classification, in the network’s own terminology) with-
 269 out meaningfully altering bank lending behavior on the ground, precisely the
 270 state-capacity/policy-credibility distinction this paper is built to test. The
 271 evidence that green finance works only through capable institutions is by

272 now reasonably direct at the country level. Shi et al. (2024), across 37 Asian
273 economies, find that green finance improves environmental outcomes where
274 control of corruption is strong and has little detectable effect where it is weak;
275 Chi and Yang (2023) find China’s green-financial-system guidelines drive
276 firms’ green transition most where state environmental governance is strong;
277 and Jia et al. (2026) find the long-run green-credit-to-sustainable-production
278 relationship across the BRICS economies is itself strengthened by governance
279 quality. Each of these is an aggregate or single-jurisdiction result, and each
280 implies a firm-level prediction—that the credit-to-green channel should re-
281 spond to policy more sharply under stronger regulatory capacity—that has
282 not been tested directly on firm-level data across adopting economies. That
283 is the prediction H3 states and this paper tests. We proxy regulatory ca-
284 pacity with the World Bank’s Worldwide Governance Indicators Regulatory
285 Quality estimate, matched to each economy’s survey year, and hypothesize
286 a three-way interaction:

287 **H3.** The policy-moderation effect described in H2 is itself in-
288 creasing in regulatory quality: SBFN adoption strengthens the
289 credit-to-green-investment channel most where the supervising
290 jurisdiction has the institutional capacity to make the underly-
291 ing guideline credible, and least—potentially not at all—where it
292 does not.

293 Hypotheses H2 and H3 are, by construction, statements about how a
294 country-level variable reshapes a firm-level relationship. That two-level struc-
295 ture is what Section 4 argues a classical single-level specification is poorly

placed to adjudicate on this sample—not because the interaction terms it tests are unidentified, but because a country-cluster count in the low forties makes cluster-robust inference on them unreliable, and because a country-fixed-effects variant would purchase within-country identification at the cost of the level-2 main effects and of resting on only eight adopter economies. That is what motivates the hierarchical and machine-learning estimators we turn to after reporting the classical benchmark.

3.4. *An auxiliary hypothesis: firm-level absorptive capacity as an alternative margin*

H2 and H3 are both, in the end, statements that a *country-level* characteristic conditions the strength of the credit-to-green channel. A separate strand of the financing-constraints literature suggests the conditioning variable that matters most may instead sit at the *firm* level: B. Ullah (2025) shows that financial constraints bind corporate environmental responsibility specifically at smaller firm scales, consistent with a standard fixed-cost logic in which the up-front capital outlay a green retrofit requires (Section 3) is proportionately heavier, and therefore harder to absorb even once financed, for a smaller firm than a larger one. We did not set out to test this as a fourth ex-ante hypothesis on a par with H1–H3; it emerged from the causal forest’s feature-importance decomposition (Section 5.4), which we report honestly here as an auxiliary, exploratory hypothesis motivated by that same fixed-cost logic rather than one we pre-specified:

H4 (auxiliary, exploratory). Whatever heterogeneity exists in the strength of the credit-to-green-adoption channel is more

320 attributable to firm size than to either country-level institutional
321 moderator (SBFN status or regulatory quality).

322 We flag its exploratory status explicitly because it changes how its evi-
323 dence should be read: Section 5.4’s support for H4 is genuine and, we think,
324 informative, but it is evidence discovered during analysis rather than con-
325 firmed against a pre-registered prediction, and the same section reports hon-
326 estly that it does not yet extend to a clean, monotonic size gradient. H4
327 also connects this paper’s evidence to the just-transition and SME-financing
328 literature’s broader concern with who is left behind when a policy instru-
329 ment operates on an extensive banking margin rather than on the firm-scale
330 margin that determines whether a financed firm can absorb a green invest-
331 ment’s fixed costs at all—a distributional dimension of green-credit-policy
332 design this paper’s data can only gesture toward, not test directly, but which
333 we think is a natural extension for future work using firm-level data with
334 a continuous size measure and a larger, more granular firm-size distribution
335 than the Enterprise Surveys’ own three-category strata provide.

336 **4. Data and methodology**

337 *4.1. Data sources and sample construction*

338 The paper’s firm-level data come from the World Bank Enterprise Sur-
339 veys (WBES), a standardized, stratified-random-sample survey of formal-
340 sector, non-agricultural, non-extractive private firms with at least five em-
341 ployees, conducted by the World Bank’s Enterprise Analysis Unit. Our pri-
342 mary analysis sample is the World Bank’s own harmonized cross-country
343 file combining the 41 Europe and Central Asia (ECA) and Middle East and

344 North Africa (MENA) economies surveyed under a single, identical question-
 345 naire between 2018 and 2020, the survey wave in which the Bank’s Green
 346 Economy module—a battery of questions on emissions monitoring, energy-
 347 consumption targets, and the adoption of specific climate-friendly production
 348 measures—was fielded in full to every respondent. This yields 28,042 firm-
 349 level observations across 41 economies (Table 2). We supplement this primary
 350 sample with 15,556 additional firm observations from six further economies
 351 (Bangladesh 2022, India 2022, Indonesia 2023, Peru 2023, Philippines 2023,
 352 Timor-Leste 2021) in which a shortened, partially randomized version of the
 353 Green Economy module was fielded; because the exact item wording and ran-
 354 domized sub-module assignment differ across these six economies, we do not
 355 pool them into the main harmonized outcome variable, and instead use them
 356 as an out-of-region external-validity check on a narrower, honestly-matched
 357 pair of indicators (CO2-emissions monitoring, available in five of the six;
 358 waste-minimization adoption, available in five of the six, with different spe-
 359 cific economies covered by each), reported separately in Section 5.

360 The primary sample, while broad by the standards of the existing green-
 361 credit-policy literature, is not the full universe of WBES economies carrying
 362 *any* Green Economy content, and we build a second, independent sample
 363 to exploit the rest of it. The World Bank also maintains a standardized
 364 cross-country database (2006–2026) that harmonizes a common set of core
 365 variables across every Formal Sector Enterprise Survey wave it has ever
 366 fielded, 292,314 firm-level observations across 471 country-years. Within
 367 it, 160 country-years fielded between 2021 and 2026 — reaching, notably,
 368 into high-income economies entirely outside the SBFN’s remit such as Aus-

369 tralia, Canada, Germany, and Japan — carry a harmonized minimal Green
 370 Economy item set: climate-damage exposure and, more relevantly for our
 371 outcome construct, CO2-emissions monitoring, alongside the same standard
 372 access-to-finance module (credit line/loan, overdraft, perceived finance ob-
 373 stacle) used in the primary sample. We supplement this with Bangladesh
 374 2022 and Indonesia 2023 — two economies whose Green Economy responses
 375 used an alternative randomized-module variable naming convention and so
 376 are absent from the standardized database’s harmonized columns, but which
 377 we can recover from our own extraction of the underlying country files —
 378 for a combined global sample of 162 economies and 100,115 firms (89,797
 379 with complete data on all covariates). Firm size, sector, and productivity
 380 controls for this sample are drawn from the Bank’s companion Productivity
 381 Database, merged by firm identifier. We refer to this as the *global sample*,
 382 and to CO2-emissions monitoring as its outcome variable throughout; it is
 383 the paper’s vehicle for testing whether the primary sample’s central find-
 384 ing replicates independently, at far greater scale, and under an identification
 385 strategy — country fixed effects — that the primary sample’s single-wave
 386 design cannot support (Section 4).

387 Country-level data are drawn from two sources for both samples. For the
 388 primary sample’s 47 economy-years, SBFN policy-adoption status, join year,
 389 and progression-framework stage as of each economy’s survey year are com-
 390 piled from the Sustainable Banking and Finance Network’s Global Progress
 391 Reports (2018, 2019, 2021, 2023–2024), country-level Progress Report ad-
 392 denda, and the network’s live country-profile data portal, cross-checked against
 393 IFC press releases announcing individual member accessions; economies that

394 joined SBFN only after their WBES survey year are coded as non-adopters
395 as of that year (Section 2). For the global sample’s remaining 157 economy-
396 years, we draw membership status and join year directly from the SBFN data
397 portal’s full published member roster (67 members as of the portal’s most
398 recent update), applying the identical ex-ante rule: an economy is coded as
399 an adopter only if its join year precedes its survey year. Six economy-years
400 in the global sample (Central African Republic, Cameroon, Chad, Republic
401 of Congo, Equatorial Guinea, and Gabon, all 2023–2024) are recorded on
402 the roster under a regional entry, “Central African States,” which we inter-
403 pret as referring to the CEMAC zone’s common banking supervisor; four
404 small-island economy-years (Antigua and Barbuda, Grenada, St. Lucia, St.
405 Vincent and the Grenadines, all 2025) are similarly recorded under “East-
406 ern Caribbean States,” which we interpret as the Eastern Caribbean Central
407 Bank’s member states. Both interpretations are flagged explicitly here as
408 inferences rather than explicit per-country listings. Regulatory capacity is
409 measured with the World Bank’s Worldwide Governance Indicators (WGI)
410 Regulatory Quality estimate, matched to each economy’s exact survey year
411 (or, where not yet published, the most recent prior year available) via the
412 World Bank DataBank API; WGI does not cover Taiwan, which is accord-
413 ingly dropped from global-sample regressions.

414 A natural alternative to the cross-sectional design used throughout this
415 paper would exploit SBFN’s own staggered adoption timing—join years in
416 our sample span 2012–2024 (Section 2)—in a staggered difference-in-differences
417 or event-study design around each economy’s own adoption date, the identi-
418 fication strategy this literature most often asks for. We do not attempt this,

and it is worth being explicit about why rather than leaving the gap implicit. The WBES Green Economy module is not a repeated panel instrument: it has been fielded to a given country, in most cases, exactly once, as a special one-off module attached to a single survey wave, and even the small subset of economies in our sample surveyed more than once by the WBES core instrument were not re-surveyed with a comparable Green Economy module in a second wave; the six supplementary economies in our own extension sample (Section 5.5) illustrate the problem directly, since even they differ from one another, and from the primary sample, in exact item wording and randomized sub-module assignment, precluding a consistently defined outcome measured twice for the same country before and after its SBFN join date. A within-country before/after comparison around each economy’s own adoption date is accordingly not implementable with a consistent outcome variable in the currently published WBES microdata, for this specific outcome construct. This is a genuine data-availability constraint on the identification strategy available to us, not a design choice; we flag it here as a first-class limitation of the paper’s identification, and return to its consequences in Section 8.

4.2. Variable construction

Outcome. Our primary dependent variable, *green practice adoption*, is a binary indicator equal to one if a firm reports having adopted at least one of seven Green Economy module practices over the preceding three years: more climate-friendly on-site energy generation, an energy-management system, waste-minimization or recycling measures, air-pollution control measures, other pollution-control measures, any energy-efficiency measure, or the use of on-site renewable energy. We also construct a continuous *green*

444 *adoption index*, the mean of the same seven binary items, for robustness.

445 **Treatment.** *Bank credit access* is a binary indicator equal to one if the
446 firm reports holding a line of credit or loan from a formal financial institution
447 at the time of the survey.

448 **Country-level moderators.** *SBFN member* is a binary indicator equal
449 to one if the economy had adopted a national SBFN sustainable-finance
450 policy framework as of the survey year. *Regulatory quality* is the WGI Reg-
451 ulatory Quality estimate for the survey year (continuous, standardized to
452 mean zero, unit standard deviation within the estimation sample for the
453 hierarchical and machine-learning specifications).

454 **Controls.** Firm size (small/medium/large, per the Enterprise Surveys’
455 own sampling strata), broad sector (manufacturing, services, construction),
456 log sales, an exporter dummy (any direct or indirect export share), and a
457 foreign-ownership dummy (any private foreign ownership share).

458 Table 1 lists every variable used in the analysis together with its exact
459 WBES question code and source, in the interest of full transparency about
460 variable construction given the multi-source nature of the merge.

461 4.3. Empirical strategy

462 We estimate the relationship between bank credit access and green prac-
463 tice adoption, and its moderation by SBFN status and regulatory quality,
464 in five stages of increasing methodological sophistication, each addressing a
465 specific limitation of the one before it.

466 **Stage 1 — classical benchmark.** We first estimate a pooled logistic

467 regression,

$$\Pr(\text{Green}_{i,c} = 1) = \Lambda\left(\beta_1 \text{Credit}_{i,c} + \beta_2 \text{SBFN}_c + \beta_3 (\text{Credit}_{i,c} \times \text{SBFN}_c) + \mathbf{X}'_{i,c} \boldsymbol{\gamma} + \delta_{s(i)}\right) \quad (1)$$

468 for firm i in country c , where $\mathbf{X}_{i,c}$ is the firm-control vector and $\delta_{s(i)}$ is a sec-
 469 tor fixed effect, with standard errors clustered by country. We deliberately
 470 do not include country fixed effects in this specification, and it is worth being
 471 precise about what such a fixed effect would and would not absorb. Because
 472 SBFN_c is constant within country in a single cross-sectional wave, a coun-
 473 try fixed effect absorbs its main effect entirely, making β_2 unidentifiable by
 474 construction. The interaction $\text{Credit}_{i,c} \times \text{SBFN}_c$, by contrast, is a firm-level
 475 regressor and remains formally identified under country fixed effects, since
 476 credit access itself varies within country; Stage 5 below relies on exactly that
 477 logic to identify the corresponding interactions on the global sample. What
 478 the primary sample cannot support is not the identification of β_3 but its pre-
 479 cision: with country fixed effects, β_3 would be identified only by the contrast
 480 between within-country credit variation in the eight SBFN-adopter economies
 481 of this sample and that in the remaining thirty-three (Section 5), while the
 482 SBFN main-effect level shift those economies exhibit—a first-order feature
 483 of the data—would be absorbed and unreportable. We therefore estimate
 484 β_2 and β_3 jointly without country fixed effects, as the classical benchmark
 485 against which Stage 2’s hierarchical alternative is compared, extended with
 486 a triple interaction for $\text{Regulatory quality}_c$ that tests H3’s three-way claim
 487 directly.

488 **Stage 2 — Bayesian hierarchical (multilevel) model.** We re-
 489 estimate the same relationship allowing country to enter as a random rather

490 than fixed effect, with a random slope on Credit:

$$\text{Green}_{i,c} \sim \text{Bernoulli}(p_{i,c}) \quad (2)$$

$$\begin{aligned} \text{logit}(p_{i,c}) = & (\beta_1 + u_{1,c}) \text{Credit}_{i,c} + \beta_2 \text{SBFN}_c + \beta_3 (\text{Credit}_{i,c} \times \text{SBFN}_c) \\ & + \beta_4 (\text{Credit}_{i,c} \times \text{RegQuality}_c) + \mathbf{X}'_{i,c} \boldsymbol{\gamma} + (\alpha + u_{0,c}) \end{aligned} \quad (3)$$

$$\begin{pmatrix} u_{0,c} \\ u_{1,c} \end{pmatrix} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}) \quad (4)$$

491 estimated via Hamiltonian Monte Carlo (the No-U-Turn Sampler, NUTS)
492 in PyMC, an open-source Python library for Bayesian statistical modeling
493 (Salvatier et al., 2016). This specification is the paper’s primary vehicle for
494 testing H2: because country enters through a variance component rather than
495 a full set of dummies, the country-level moderators SBFN_c and RegQuality_c
496 retain identified main effects, while the random slope $u_{1,c}$ still absorbs unob-
497 served country heterogeneity in the credit-green relationship and, together
498 with partial pooling across countries, gives more defensible inference than
499 cluster-robust standard errors computed over only 41 clusters (Cameron and
500 Miller, 2015). It carries two cross-level interactions: β_3 tests H2 directly, and
501 β_4 tests whether regulatory quality on its own moderates the credit-green
502 relationship. We are explicit that β_4 is not a test of H3. H3 asks whether
503 the SBFN moderation in H2 is itself conditional on regulatory quality, which
504 is a three-way claim requiring a $\text{Credit}_{i,c} \times \text{SBFN}_c \times \text{RegQuality}_c$ term; this
505 specification does not include one, and we do not add it here because the pri-
506 mary sample’s eight adopter economies cannot support a credibly estimated
507 three-way interaction under partial pooling. H3 is instead tested directly
508 by Stage 1’s fully saturated triple interaction above (Table 3, column M3)

509 and, on an independent sample, by Stage 5’s no-fixed-effects specification
510 (Section 5.7), both of which retain that term.

511 **Stage 3 — causal forest / double machine learning.** To avoid
512 imposing a linear interaction functional form on how the treatment effect
513 of credit access varies across the institutional landscape, we additionally es-
514 timate a Causal Forest with Double Machine Learning (Wager and Athey,
515 2018; Chernozhukov et al., 2018; Athey et al., 2019), using gradient-boosted
516 nuisance models for the outcome and treatment propensity, and firm size, ex-
517 port status, SBFN status, and regulatory quality as effect-modifying features.
518 This lets us recover conditional average treatment effects (CATEs)—the esti-
519 mated effect of credit access on green adoption, separately for SBFN members
520 versus non-members, and separately by regulatory-quality tercile—without
521 imposing that the moderation take the single-coefficient interaction form as-
522 sumed in Stages 1–2. We use “causal” here in the qualified sense standard
523 in this literature: the forest’s identifying assumption is unconfoundedness
524 conditional on the observed covariate set (firm size, export status, SBFN
525 status, regulatory quality), not an exogenous source of variation in credit
526 access itself. This is exactly the same assumption underlying the logit and
527 hierarchical specifications; Stage 3 relaxes the functional form imposed on
528 that assumption from linear to fully flexible, but it does not relax the as-
529 sumption, and the CATEs reported below should be read accordingly as
530 doubly-robust conditional associations rather than causal effects identified
531 off a natural experiment.

532 **Stage 4 — small-sample supplementary check.** We additionally
533 re-estimate a reduced-form version of the credit-green relationship using a

waste-minimization-adoption outcome, available in four economies (India 2022, Indonesia 2023, Peru 2023, Timor-Leste 2021) whose Green Economy modules asked this item under directly comparable wording. This outcome is not part of the standardized cross-country database underlying Stage 5 below, so it offers genuinely independent, if small-sample, supplementary evidence.

Stage 5 — global-sample replication with country fixed effects.

Finally, and centrally, we re-estimate the full specification on the 162-economy global sample described in Section 4, using CO2-emissions monitoring as the outcome. Because this sample spans far more country clusters than the primary sample, we can now afford a country-fixed-effects specification: δ_c absorbs SBFN_c and RegQuality_c 's main effects (collinear with the fixed effect by construction, since both are constant within country-year) while leaving their firm-level interactions with $\text{Credit}_{i,c}$ fully identified, since credit access itself varies within country. This is a materially sharper test of H2 than Stage 1's clustered-SE approach, on an entirely independent sample and outcome from Stages 1–3. The fixed-effects specification carries the two two-way interactions only, so we report it alongside a no-fixed-effects specification for two reasons: direct comparability with the primary sample's Table 3, and because that specification, like Table 3's M3, retains the triple interaction H3 requires.

5. Results

5.1. Descriptive patterns

Table 1: Variable definitions and sources.

Variable	Definition	WBES code	Source
green_adoption_binary	=1 if firm adopted ≥ 1 of 7 Green Economy practices in last 3 years	BMGc23b, c23d, c23e, c23f, c23j, c25, e5	WBES Green Economy Module (2018–2020)
fin_has_credit_line	=1 if firm has a line of credit or loan	K8	WBES Access to Finance Module
sbfn_member	=1 if SBFN policy adopted as of survey year	n/a	SBFN Global Progress Reports; data.sbfnetwork.org
wgi_regulatory_quality	WGI Regulatory Quality estimate, survey year	n/a	World Bank DataBank API (GOV_WGI_RQ_EST)
size_cat	Firm size stratum (Small/Medium/Large)	A6A	WBES Sampling Frame
sector_broad	Broad sector (Manufacturing/Services/Construction)	A4AX	WBES Sampling Frame
log_sales	Log of total annual sales	D2	WBES General Information Module
foreign_owned_dummy	=1 if any foreign ownership share $>0\%$	B2B	WBES Ownership Module
exporter_dummy	=1 if any export sales share $>0\%$	D3B, D3C	WBES Sales Module

Table 2: Sample composition, by economy.

Economy	<i>N</i> firms	Year	SBFN	Reg. qual.	Green rate	Credit rate
Albania 2019	377	2019	0	0.09	0.609	0.407
Armenia 2020	546	2020	0	0.21	0.684	0.482
Azerbaijan 2019	225	2019	0	-0.10	0.418	0.180
Belarus 2018	600	2018	0	-0.79	0.662	0.425
Bosnia and Herze- govina 2019	362	2019	0	-0.38	0.581	0.565
Bulgaria 2019	772	2019	0	0.46	0.604	0.467
Croatia 2019	404	2019	0	0.38	0.708	0.448
Cyprus 2019	240	2019	0	1.04	0.746	0.711
Czechia 2019	502	2019	0	1.12	0.725	0.526
Egypt 2020	3,075	2020	1	-0.64	0.303	0.065
Estonia 2019	360	2019	0	1.54	0.738	0.412
Georgia 2019	581	2019	1	0.88	0.441	0.501
Greece 2018	600	2018	0	0.29	0.825	0.499
Hungary 2019	805	2019	0	0.46	0.673	0.537
Italy 2019	760	2019	0	0.70	0.506	0.230
Jordan 2019	601	2019	1	0.16	0.471	0.248
Kazakhstan 2019	1,446	2019	0	0.11	0.519	0.257
Kosovo 2019	271	2019	0	-0.38	0.698	0.425
Kyrgyz Republic 2019	360	2019	1	-0.42	0.606	0.299
Latvia 2019	359	2019	0	1.00	0.824	0.386

Economy	N firms	Year	SBFN	Reg. qual.	Green rate	Credit rate
Lebanon 2019	532	2019	0	-0.56	0.452	0.365
Lithuania 2019	358	2019	0	0.98	0.678	0.342
Malta 2019	242	2019	0	0.83	0.901	0.479
Moldova 2019	360	2019	0	-0.11	0.619	0.328
Mongolia 2019	360	2019	1	-0.07	0.625	0.492
Montenegro 2019	150	2019	0	0.29	0.387	0.576
Morocco 2019	1,096	2019	1	0.03	0.478	0.251
North Macedonia 2019	360	2019	0	0.19	0.572	0.508
Poland 2019	1,369	2019	0	0.83	0.569	0.403
Portugal 2019	1,062	2019	0	1.00	0.812	0.600
Romania 2019	814	2019	0	0.36	0.438	0.474
Russia 2019	1,323	2019	0	-0.35	0.503	0.310
Serbia 2019	361	2019	0	0.06	0.654	0.585
Slovak Republic 2019	429	2019	0	0.73	0.782	0.386
Slovenia 2019	409	2019	0	0.88	0.793	0.674
Tajikistan 2019	352	2019	0	-0.80	0.523	0.194
Tunisia 2020	615	2020	1	-0.11	0.391	0.413
Türkiye 2019	1,663	2019	1	-0.02	0.346	0.345
Ukraine 2019	1,337	2019	0	-0.29	0.715	0.244
Uzbekistan 2019	1,239	2019	0	-0.92	0.751	0.240

Economy	<i>N</i> firms	Year	SBFN	Reg. qual.	Green rate	Credit rate
West Bank and Gaza 2019	365	2019	0	-0.15	0.428	0.178

Note: Reg. qual. is the WGI Regulatory Quality estimate for the survey year. Green rate is the unconditional share of firms reporting adoption of at least one Green Economy practice. Credit rate is the share of firms reporting a bank credit line or loan.

557 Table 2 reports the full 41-economy sample composition; firm counts range
558 from 150 (Montenegro) to 3,075 (Egypt), and the raw, unconditional green-
559 practice-adoption rate ranges more than three-fold across economies, from
560 30.3% (Egypt) to 90.1% (Malta). Figure 1 maps this variation alongside
561 SBFN policy status: 8 of the 41 primary-sample economies had adopted
562 an SBFN framework as of their survey year (Georgia, Egypt, Jordan, Kyr-
563 gyz Republic, Mongolia, Morocco, Tunisia, and Türkiye)—13 of the full 47-
564 economy sample once the five (of six) SBFN-member small-sample supple-
565 mentary economies are counted—and every European Union member state in
566 the sample is, consistent with SBFN’s mandate, a non-adopter. At the firm
567 level, bank credit access correlates positively with the green-adoption index
568 ($r = 0.18$) but is essentially uncorrelated with a firm’s own perceived finance
569 obstacle ($r = -0.02$)—a first hint, well before any regression, that realized
570 access to credit rather than perceived financing difficulty is what tracks green
571 behavior, and the two are evidently not interchangeable measures of the same
572 underlying constraint.

573 Figure 2 previews the paper’s central finding graphically before a sin-
574 gle regression is estimated: plotting each economy’s green-adoption rate

575 against its regulatory-quality score and fitting separate lines for SBFN mem-
 576 bers and non-members produces two lines of very similar, modestly positive
 577 slope, separated by a roughly 15–20 percentage-point vertical gap—SBFN-
 578 member economies show systematically *lower* raw green-adoption rates at
 579 every level of regulatory quality, not a steeper relationship between regu-
 580 latory quality and adoption. That combination—a level difference with no
 581 slope difference—is exactly the signature the regression evidence below con-
 582 firms formally.

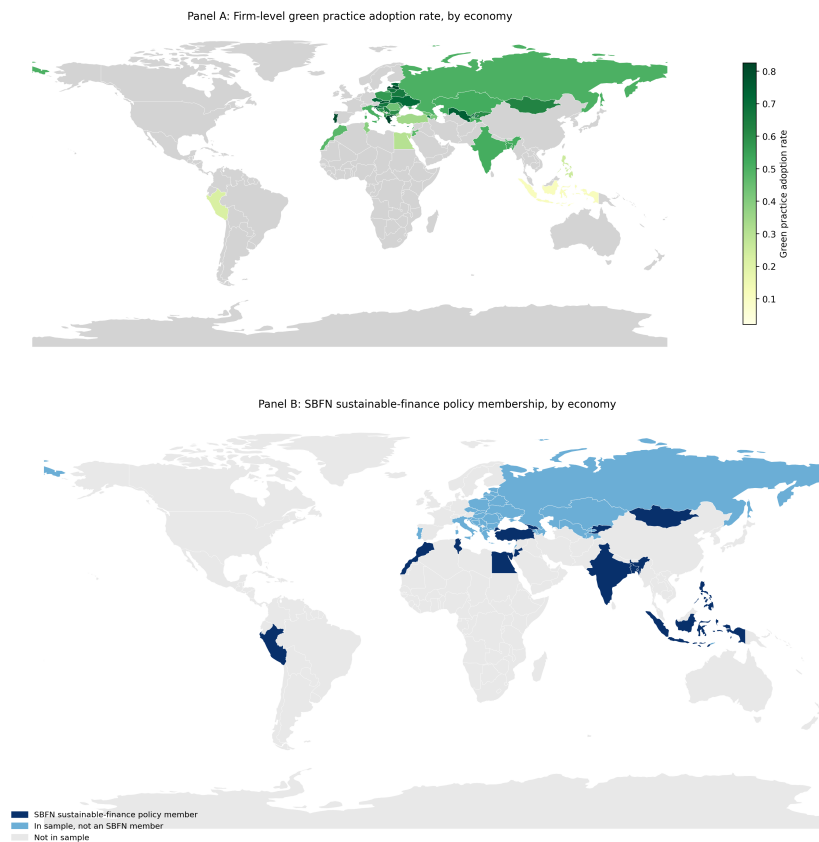


Figure 1: Green practice adoption and SBFN policy membership, by economy.

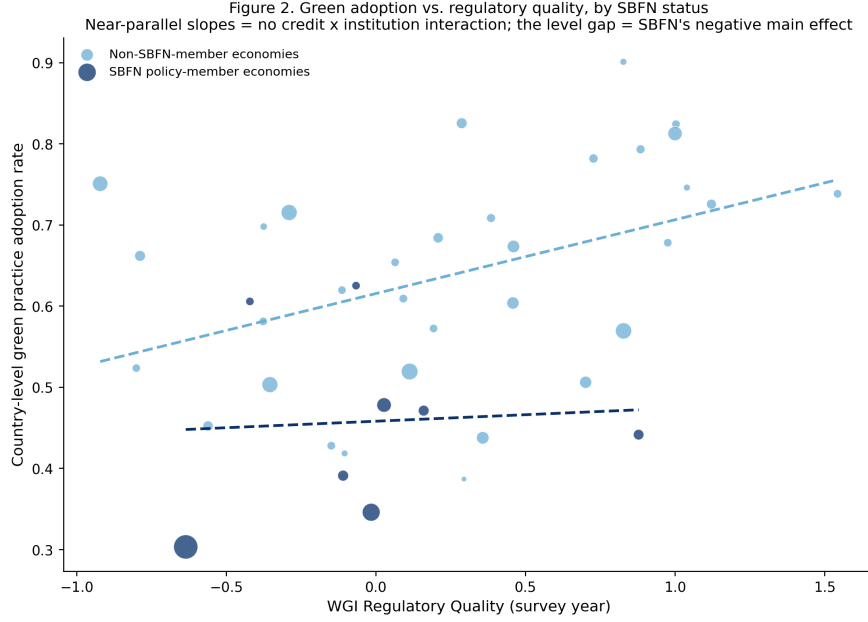


Figure 2: Green adoption vs. regulatory quality, by SBFN status.

5.2. Baseline classical benchmark

Table 3 reports the pooled logistic baseline. Column M1 confirms H1 on its own terms: firms with a bank credit line are significantly more likely to have adopted a green practice ($b = 0.571$, $p < 0.001$; average marginal effect = 12.95 percentage points, 95% CI [7.7, 18.2])—holding a formal credit line is associated with roughly the same increase in the probability of green adoption as moving from a small to a large firm, and a somewhat larger increase than being an exporter. Column M2 adds SBFN status and its interaction with credit access: the SBFN main effect is large, negative, and precisely estimated ($b = -0.991$, $p < 0.001$), while the credit $\times$ SBFN interaction is small and statistically indistinguishable from zero ($b = 0.049$, $p = 0.773$). Column M3 adds the fully saturated triple interaction with regulatory quality, and

595 it is this column, not the hierarchical model of Section 5.3, that carries the
596 paper’s direct test of H3: the credit $\times$ SBFN $\times$ regulatory-quality term is
597 small, wrong-signed relative to H3’s prediction, and nowhere near significance
598 ($b = -0.104$, $se = 0.272$, $p = 0.702$). The rest of M3 tells the same story
599 at greater length: every interaction term involving credit access is small
600 and insignificant, while regulatory quality’s uninteracted relationship with
601 adoption is positive, if only marginally significant in this particular fully-
602 saturated specification ($b = 0.278$, $p = 0.110$).

603 As set out in Section 4, adding country fixed effects here would absorb the
604 SBFN and regulatory-quality main effects entirely, while leaving the interac-
605 tion terms identified but reliant on the contrast between within-country credit
606 variation in this sample’s eight adopter economies and its thirty-three non-
607 adopters. Table 3 accordingly reports main effects and interactions jointly,
608 without country fixed effects; what it shows is a genuine test of H2 and, in
609 M3, of H3, not a placeholder—and that test returns no interaction on ei-
610 ther. The question the remaining two stages are built to answer is whether
611 that null result reflects a real absence of policy moderation, or an artifact
612 of forcing a two-level question through a single-level model with country-
613 clustered standard errors computed over only 41 clusters, a setting in which
614 such standard errors are known to be unreliable (Cameron and Miller, 2015).

615 5.3. Bayesian hierarchical model — primary test of H2

616 The hierarchical specification—country-varying random slopes on credit
617 access, cross-level interactions with SBFN status and standardized regulatory
618 quality, four chains of 1,000 post-warmup draws each, all $\hat{R} \leq 1.01$ and
619 ess_{bulk} between 797 and 2,270—is built specifically to give H2 a fair hearing

Table 3: Baseline classical logit regressions (dependent variable: green practice adoption, binary). Country-clustered standard errors in parentheses.

	M1	M2	M3
Credit access	0.571*** (0.123)	0.459*** (0.090)	0.373*** (0.085)
SBFN member		-0.991*** (0.178)	-0.844*** (0.209)
Credit $\times$ SBFN		0.049 (0.168)	0.019 (0.140)
Regulatory quality			0.278 (0.174)
Credit $\times$ Reg. quality			0.144 (0.142)
SBFN $\times$ Reg. quality			0.053 (0.261)
Credit $\times$ SBFN $\times$ Reg. quality			-0.104 (0.272)
Log sales	0.061 (0.039)	0.041 (0.034)	0.068** (0.032)
Foreign owned	0.219 (0.143)	0.244* (0.139)	0.220 (0.143)
Exporter	0.537*** (0.092)	0.471*** (0.074)	0.412*** (0.079)
Sector, size controls	Yes	Yes	Yes
N	23,914	23,914	23,914
Pseudo R^2	0.058	0.089	0.093

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

under inference the 41-cluster classical specification cannot deliver, and to test regulatory quality’s own two-way moderation alongside it. H3’s three-way claim is not tested here (Section 4); it is tested in Table 3’s M3 above and in Section 5.7 below, and returns a null in both. This specification does not change the conclusion either. Table 4 reports the full posterior summary; three results stand out.

First, the firm-level credit effect (H1) survives unchanged in substance: posterior mean 0.33 (89% equal-tailed interval [0.18, 0.49]), comfortably excluding zero. Second, both country-level institutional variables retain clean, precisely estimated *main* effects on the baseline adoption rate—SBFN membership net-negative (-0.57, 89% ETI [-1.00, -0.09]), regulatory quality net-positive (0.34, 89% ETI [0.16, 0.53])—reproducing, with proper hierarchical uncertainty quantification rather than an absorbed or clustered approximation, exactly the level-shift pattern Figure 2 shows visually. Third, and centrally for H2, neither cross-level interaction is distinguishable from zero: credit $\times$ SBFN, 0.072 (89% ETI [-0.26, 0.40]); credit $\times$ regulatory quality (standardized), 0.004 (89% ETI [-0.13, 0.14])—a near-exact zero on the latter, not merely an imprecise one. The first of these is the direct H2 test; the second establishes that regulatory quality does not moderate the credit-green channel on its own, which is a necessary condition for the mechanism H3 posits without being a test of H3 itself.

We read this as an honest and, we think, informative finding rather than a disappointing one: giving the moderation hypothesis a hierarchical model built for precisely this two-level question still returns no interaction, while sharpening (relative to the pooled baseline) the confidence with which we

can state that SBFN status and regulatory quality shift the *level* of green adoption directly.

Table 4: Bayesian hierarchical model, posterior summary (four chains, 1,000 post-warmup draws each).

Parameter	Mean	SD	89% ETI low	89% ETI high	$\hat{R}$
Intercept	-1.93	0.246	-2.30	-1.50	1.00
Credit access	0.33	0.097	0.18	0.49	1.00
SBFN member	-0.57	0.300	-1.00	-0.09	1.01
Credit $\times$ SBFN	0.072	0.211	-0.26	0.40	1.00
Reg. quality (std.)	0.344	0.116	0.16	0.53	1.00
Credit $\times$ Reg. quality	0.004	0.087	-0.13	0.14	1.00

5.4. Causal forest / heterogeneous treatment effects

Table 5 and Figure 5 report the Causal Forest DML estimates, which impose no linear-interaction functional form and let the treatment effect vary flexibly across firm and country covariates. The overall average treatment effect of credit access is 0.073 (95% CI [-0.040, 0.186])—directionally consistent with, though smaller in magnitude and less precisely estimated than, the logit average marginal effect of 0.130 in Section 5.2, reflecting the more conservative variance properties of a doubly-robust, cross-fitted nonparametric estimator relative to a parametric logit on the same data. Critically, the conditional average treatment effects triangulate with Section 5.3 rather than contradicting it: the estimated effect is 0.072 for firms in non-SBFN economies versus 0.076 for firms in SBFN economies, and 0.075 / 0.070 /

0.075 across the low/middle/high regulatory-quality terciles respectively—
three different institutional cuts, three essentially flat effects, all visibly overlapping in Figure 5. Two entirely independent estimators, one parametric and hierarchical, one nonparametric and forest-based, agree on the absence of institutional moderation.

The forest’s feature-importance decomposition adds a genuinely new result rather than only confirming the null, and speaks directly to the auxiliary H4 introduced in Section 3.4: of the four candidate effect-modifiers, firm size (log sales) accounts for 75.6% of the model’s explained heterogeneity, versus 19.9% for regulatory quality, 2.6% for SBFN status, and 1.9% for exporter status. Whatever heterogeneity exists in how strongly credit access predicts green adoption is, on this evidence, primarily a firm-level rather than a country-institutional phenomenon, supporting H4. Collapsing this into the Enterprise Surveys’ own coarse size strata, however, does not reveal a clean monotonic gradient: the estimated effect is 0.080 for small firms, 0.075 for medium firms, and 0.057 for large firms (Table 6; all with wide, overlapping confidence intervals spanning zero), which if anything runs opposite to a simple “larger firms better absorb a green investment’s fixed costs” story rather than confirming it. Read alongside the 75.6% importance share for the continuous size measure, we take this as evidence that the heterogeneity the forest is detecting is more granular or non-monotonic across the firm-size distribution than a three-category collapse can reveal, rather than evidence for any particular direction of a size gradient—a qualification we report explicitly rather than paper over with a tidier-sounding but unsupported claim. This redirects, rather than closes, the paper’s institutional-moderation question,

684 and we return to it in Section 7.

Table 5: Causal forest DML: average treatment effect, CATEs, and feature importances.

Estimate	Value	95% CI low	95% CI high
Overall ATE	0.0733	-0.0396	0.1862
CATE SBFN non-member	0.0723	-0.0300	0.1746
CATE SBFN member	0.0758	-0.0591	0.2107
CATE reg. quality tercile: low	0.0753	-0.0568	0.2075
CATE reg. quality tercile: mid	0.0697	-0.0355	0.1749
CATE reg. quality tercile: high	0.0749	-0.0203	0.1701
Feature	Importance		
SBFN member	0.0257		
Regulatory quality	0.1993		
Log sales	0.7562		
Exporter dummy	0.0189		

685 5.5. Small-sample supplementary check: waste-minimization adoption

686 Before turning to the global-sample replication that is this paper’s second
687 major test (Section 5.7), we report a smaller supplementary check using
688 waste-minimization adoption—an outcome not captured in the standardized
689 cross-country database underlying the global sample, and so a genuinely
690 independent (if small-sample) data point—available in four economies whose
691 Green Economy modules asked this item under directly comparable wording:
692 India 2022, Indonesia 2023, Peru 2023, and Timor-Leste 2021 (descriptive
693 rates in Supplementary Table S1). Three of these four are themselves SBFN

Table 6: Causal forest CATEs by firm-size category.

Size category	CATE	95% CI low	95% CI high
Small	0.0797	-0.0409	0.2004
Medium	0.0747	-0.0344	0.1838
Large	0.0569	-0.0428	0.1565

members, so this sample cannot re-test the SBFN-adoption contrast; what it can test is whether the credit-to-green channel documented in Sections 5.2–5.4 generalizes beyond the ECA-MENA sample. It does: pooled across the four economies, having an overdraft facility is associated with significantly higher odds of waste-minimization adoption ($b = 0.422$, $p < 0.001$, $N = 9,038$, 4 countries)—a preview, on a different continent and half-decade, of the much larger replication in Section 5.7.

5.6. Additional robustness

Supplementary Table S2 subjects the primary-sample finding to four further checks (sector fixed effects, firm controls, country-clustered standard errors; no country fixed effects, for the reason given in Section 4). Replacing the binary outcome with the continuous seven-item green-adoption index reproduces the same pattern by OLS. Replacing bank credit access with two alternative finance measures produces one point of nuance: using overdraft-facility access, the overdraft $\times$ SBFN interaction is negative and significant ($b = -0.346$, $p = 0.031$)—the one interaction term in the paper that clears a conventional significance threshold, and it runs opposite to H2’s predicted direction—whereas using the reverse-coded perceived finance-obstacle mea-

712 sure, both the main effect and its SBFN interaction are precisely zero. Split-
 713 ting the sample into manufacturing and services subsamples shows the credit-
 714 access main effect and the null credit $\times$ SBFN interaction replicate within
 715 each sector separately. We read the overdraft result as suggestive rather than
 716 conclusive: it is one significant coefficient among more than a dozen interac-
 717 tion tests across the paper, and a Benjamini–Hochberg false-discovery-rate
 718 correction applied to that full family (Supplementary Table S3) pushes its
 719 raw $p = 0.031$ to $p = 0.403$ —no coefficient in the family survives at a false
 720 discovery rate of 0.10 or even 0.40, exactly what a pure false-discovery process
 721 would be expected to produce roughly one time in twenty.

722 5.7. *Global-sample replication with country fixed effects*

723 The preceding five subsections establish the paper’s central finding on the
 724 primary sample. We now subject that finding to what we consider the most
 725 demanding test available to us: complete re-estimation on an independently
 726 assembled, 162-economy global sample (Section 4), using a different outcome
 727 variable (CO2-emissions monitoring rather than the seven-item composite),
 728 different survey years (2021–2026 rather than 2018–2020), and, for the first
 729 time in the paper, a country-fixed-effects specification that this much larger
 730 set of clusters can support. Figure 3 maps the combined reach of the two
 731 samples together: 166 distinct economies, spanning every region and, through
 732 the global sample specifically, extending for the first time in this literature
 733 into high-income economies with no SBFN involvement at all.

734 Table 7 reports two specifications. Column M1, without country fixed ef-
 735 fects, is the direct global-sample analogue of Table 3’s primary-sample base-
 736 line: credit access remains positive and significant ($b = 0.198$, $p = 0.039$,

⁷³⁷ $N = 89,797$, 159 countries), while the SBFN main effect ($b = -0.171$,
⁷³⁸ $p = 0.353$), the credit $\times$ SBFN interaction ($b = 0.278$, $p = 0.430$), and
⁷³⁹ every regulatory-quality interaction are statistically indistinguishable from
⁷⁴⁰ zero. Like Table 3's M3, this specification is fully saturated and so retains
⁷⁴¹ the credit $\times$ SBFN $\times$ regulatory-quality term H3 requires, which is likewise
⁷⁴² insignificant; Table 7 tabulates the focal two-way terms only. H3 therefore re-
⁷⁴³ ceives a direct, fully saturated test on both firm-level samples, and returns a
⁷⁴⁴ null on each. Column M2 imposes country fixed effects, absorbing the SBFN
⁷⁴⁵ and regulatory-quality main effects by construction (they are collinear with
⁷⁴⁶ δ_c , since both are constant within a given country-year) while leaving their
⁷⁴⁷ firm-level interactions with credit access fully identified, because credit access
⁷⁴⁸ itself varies within country. This is a sharper test than anything the primary
⁷⁴⁹ sample's single 2018–2020 wave can support, and it returns the same answer
⁷⁵⁰ with even more precision: credit access is strongly significant ($b = 0.249$,
⁷⁵¹ $p < 0.001$), while the credit $\times$ SBFN interaction ($b = 0.506$, $p = 0.205$) and
⁷⁵² the credit $\times$ regulatory-quality interaction ($b = -0.044$, $p = 0.346$) remain
⁷⁵³ null.

⁷⁵⁴ We regard this as the paper's most important robustness result. It is not
⁷⁵⁵ a re-analysis of the same data under a different label: it is a different outcome
⁷⁵⁶ construct, a different set of economies (only a handful of which overlap with
⁷⁵⁷ the primary sample), a different half-decade, and an econometric specification
⁷⁵⁸ — country fixed effects — that is only available at all because this second
⁷⁵⁹ sample has enough clusters to support it. That the qualitative conclusion
⁷⁶⁰ is identical across both samples and across the full set of four estimators
⁷⁶¹ reported in this paper (classical logit, Bayesian hierarchical, causal forest, and

now country-fixed-effects logit) is, we think, considerably stronger evidence
for the paper’s central claim than either sample could offer alone.

Figure 3. Combined sample coverage: 166 economies (41 primary rich-module + 162 global minimal-indicator, net of overlap)

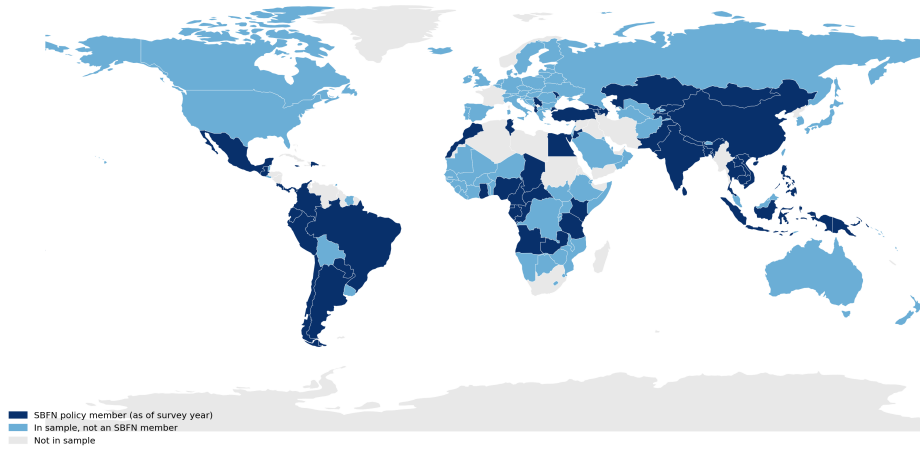


Figure 3: Combined sample coverage: 166 economies (41 primary + 162 global, net of overlap), shaded by SBFN status.

5.8. Results summary across all specifications

Sections 5.2–5.7 report ten tables spanning four estimators and two independent samples. Figure 4 consolidates every coefficient bearing on H1–H3 into a single view. Panel A shows the credit-access effect (H1): every point estimate is positive and every confidence interval excludes zero, in both samples and all four estimators. Panels B and C show the credit $\times$ SBFN and credit $\times$ regulatory-quality interactions (H2, H3): every single confidence interval, across both samples and every estimator, straddles zero. Figure 5 shows the same non-result from a completely different angle—the causal forest’s conditional average treatment effects broken out by SBFN status, regulatory-quality tercile, and firm-size category all overlap both each other

Table 7: Global-sample regressions (dependent variable: CO2-emissions monitoring, binary). Country-clustered standard errors in parentheses; M2's SBFN and regulatory-quality main effects are absorbed by country fixed effects and are not separately estimable.

	M1: no country FE	M2: country FE
Credit access	0.198** (0.096)	0.249*** (0.057)
SBFN member	-0.171 (0.184)	— (absorbed)
Credit $\times$ SBFN	0.278 (0.352)	0.506 (0.399)
Regulatory quality (std.)	0.155 (0.096)	— (absorbed)
Credit $\times$ Reg. quality	0.092 (0.089)	-0.044 (0.047)
Log sales	0.127*** (0.027)	0.288*** (0.028)
Foreign owned	0.621*** (0.083)	0.528*** (0.046)
Exporter	0.456*** (0.094)	0.335*** (0.044)
Country fixed effects	No	Yes
N	89,797	89,797
Countries	159	159

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Taiwan dropped (no WGI coverage).

and the overall average treatment effect. No subgroup split, by any variable examined in this paper, recovers a distinguishable effect.

Figure 4. Coefficient estimates across every specification and both samples
(Panel A: robust and significant everywhere; Panels B-C: null everywhere)

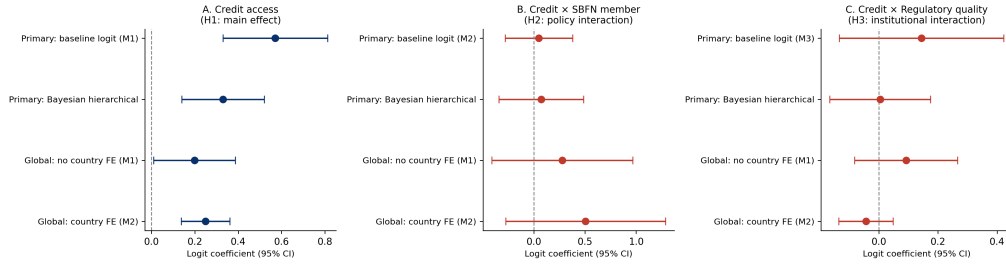


Figure 4: Coefficient estimates across every specification and both samples.

6. Evaluating the aggregate outcome: a staggered-adoption event study

Section 5 evaluates link 4 of the policy’s theory of change (Section 2.1) using firm-level cross-sectional data, for the data-availability reasons given in Section 4: the WBES Green Economy module cannot support a staggered difference-in-differences or event-study design around each economy’s own SBFN adoption date, because it has not been fielded as a repeated panel instrument. That constraint does not apply to link 5, the aggregate outcome the policy is ultimately meant to move. The same SBFN adoption-timing variation can answer it directly: does a country’s environmental trajectory shift after it joins? This section reports that test, using an independent data source and unit of analysis and the impact-evaluation design best suited to staggered policy adoption.

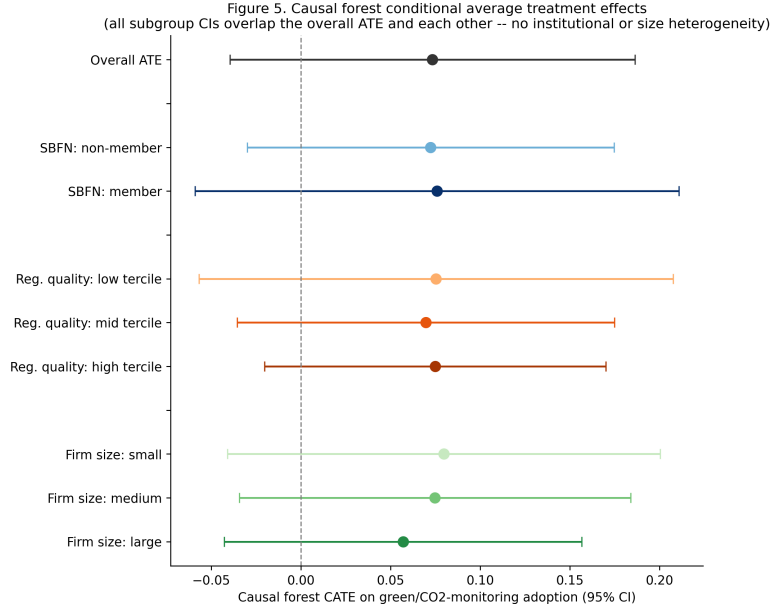


Figure 5: Causal forest conditional average treatment effects, by subgroup.

6.1. Data and design

We assemble a country-year panel spanning every World Bank member economy with a non-aggregate ISO3 code (217 economies), 2000–2024. Treatment timing is each economy’s SBFN join year, drawn from the same SBFN data portal membership roster used to construct the global sample (Section 4) and cross-checked against the global-sample SBFN coding used in Section 5.7 (zero disagreements across 61 overlapping economies); economies that have never joined SBFN, as of the roster’s most recent update, serve as the never-treated comparison group. This yields 68 ever-treated economies (join years 2012–2024, the same real staggered-adoption variation described in Section 2) and 149 never-treated economies. Outcome data come from the World Bank’s World Development Indicators, retrieved via the World

802 Bank DataBank API: renewable energy consumption (% of total final en-
803 ergy consumption, `EG.FEC.RNEW.ZS`) and CO2 emissions per capita (AR5
804 GWP-consistent series, `EN.GHG.CO2.PC.CE.AR5`), both available for the large
805 majority of economies from 2000 through at least 2021–2024. We addition-
806 ally draw on GDP per capita (constant 2015 US\$, `NY.GDP.PCAP.KD`) as a
807 covariate, to net out the obvious confound that SBFN-adopting economies
808 are disproportionately still-industrializing, faster-growing economies whose
809 emissions profiles are shaped by their development stage independently of
810 any banking-sector policy.

811 We estimate the effect of SBFN adoption on each outcome using the
812 Callaway and Sant’Anna (2021) staggered-adoption difference-in-differences
813 estimator (Callaway and Sant’Anna, 2021), which recovers group-time av-
814 erage treatment effects for each adoption cohort separately and aggregates
815 them into an overall average treatment effect on the treated (ATT) and
816 an event-study profile. We report this estimator, rather than a standard
817 two-way-fixed-effects (TWFE) regression, because TWFE is now well known
818 to produce badly biased estimates under staggered treatment timing when
819 already-treated units are implicitly used as controls for later-treated units
820 (Goodman-Bacon, 2021); we report a naive TWFE specification alongside
821 the Callaway-Sant’Anna estimates specifically to illustrate the size of that
822 bias in our own data, rather than as a candidate preferred estimate. For
823 each outcome we report both an unconditional specification and a doubly-
824 robust specification conditioning on log GDP per capita, and we check the
825 doubly-robust specification’s sensitivity to the choice of comparison group,
826 using never-treated economies as our preferred comparison group throughout

827 and not-yet-treated economies (those that will eventually adopt but had not
828 yet, as of the relevant relative period) as a robustness check.

829 6.2. Results

830 Table 8 reports the overall ATT for both outcomes across four specifica-
831 tions, including a comparison-group robustness check; Figure 6 plots the full
832 event-study profile (doubly-robust, GDP-per-capita-controlled, never-treated
833 specification) for both outcomes.

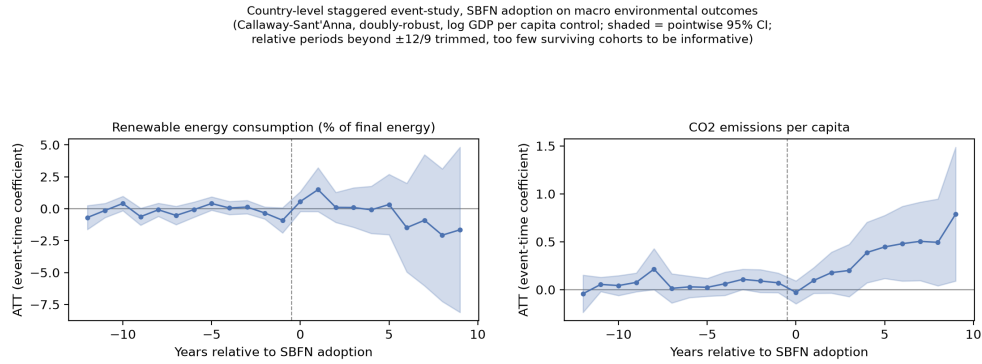


Figure 6: Country-level staggered event-study, doubly-robust specification with log GDP per capita control. Shaded bands are pointwise 95% confidence intervals; relative periods beyond $\pm 12/9$ are trimmed for having too few surviving cohorts to be informative.

834 Two results, read together, do not support treating SBFN adoption as a
835 country-level lever that visibly improves aggregate environmental outcomes –
836 which is itself informative, since this is the identification strategy best suited
837 to answer that specific question directly. First, renewable energy share shows
838 no effect distinguishable from zero in either specification, and the doubly-
839 robust event-study profile (Figure 6, left panel) is flat and precisely estimated
840 on both sides of the adoption date: a clean null, not an underpowered one.

Table 8: Country-level staggered event-study: overall average treatment effect of SBFN adoption on macro environmental outcomes. Callaway-Sant’Anna estimator, 999 bootstrap iterations; naive TWFE reported for comparison only. Full country-year panel, 2000–2024.

Outcome	Specification	ATT	SE	95% CI
Renewable energy %	Callaway-Sant’Anna, no covariates, never-treated	-1.192	0.802	[-2.763, 0.380]
Renewable energy %	Callaway-Sant’Anna, doubly-robust, log GDP p.c., never-treated	0.674	0.916	[-1.122, 2.470]
Renewable energy %	Callaway-Sant’Anna, doubly-robust, log GDP p.c., not-yet-treated	0.653	0.919	[-1.147, 2.454]
Renewable energy %	Naive TWFE (post dummy, country+year FE)	-4.672***	1.127	[-6.882, -2.463]
CO2 per capita	Callaway-Sant’Anna, no covariates, never-treated	0.631***	0.179	[0.280, 0.982]
CO2 per capita	Callaway-Sant’Anna, doubly-robust, log GDP p.c., never-treated	0.349**	0.156	[0.044, 0.655]
CO2 per capita	Callaway-Sant’Anna, doubly-robust, log GDP p.c., not-yet-treated	0.313**	0.148	[0.023, 0.603]
CO2 per capita	Naive TWFE (post dummy, country+year FE)	0.901***	0.204	[0.500, 1.302]

*** 95% CI excludes zero; ** 95% CI excludes zero, narrower margin. Underlying data and code in the replication package.

841 Second, CO2 emissions per capita shows a positive (i.e., the “wrong-
 842 signed” direction if SBFN adoption reduced emissions) and statistically sig-
 843 nificant association with adoption in every specification, but three features
 844 of the result argue against reading it causally. It shrinks by roughly 45%
 845 (0.631 to 0.349) once we condition on log GDP per capita, indicating that
 846 a substantial share of the raw association is attributable to exactly the
 847 development-stage confound we set out to net out. The doubly-robust event-
 848 study profile (Figure 6, right panel) shows a small, mostly flat pre-period
 849 with only two marginally significant leads, followed by a post-period effect
 850 that grows smoothly and monotonically from near zero at adoption to its
 851 largest magnitude nine to twelve years out – a continuing-divergence pattern
 852 more consistent with SBFN-adopting economies already sitting on a different,
 853 faster-rising emissions trajectory than non-adopters (which a single covariate
 854 cannot fully absorb) than with a discrete policy-caused jump at the adop-
 855 tion date. And the naive TWFE estimate (0.901) is itself more than double
 856 the doubly-robust Callaway-Sant’Anna estimate (0.349), the same direction
 857 of bias documented in the staggered-DID literature (Goodman-Bacon, 2021)
 858 and a useful internal illustration of why we lead with the more robust esti-
 859 mator throughout this section.

860 Both doubly-robust results are stable to the choice of comparison group:
 861 substituting not-yet-treated economies for never-treated economies leaves the
 862 CO2 estimate essentially unchanged (0.313 versus 0.349, both significant at
 863 conventional levels) and the renewable-energy null equally unchanged (0.653
 864 versus 0.674, neither distinguishable from zero); Table 8 reports both. Nei-
 865 ther conclusion in this section depends on which set of untreated economies

866 serves as the counterfactual.

867 Neither result supports a “green banking policy backfires” reading, and
868 neither supports a “green banking policy works” one. The more measured
869 reading is that a macro-level design – built using exactly the identification
870 strategy (staggered adoption timing, a proper Callaway-Sant’Anna estima-
871 tor, never-treated controls) this literature’s own methodological expectations
872 call for – also does not produce clean evidence that SBFN adoption changes
873 a country’s aggregate environmental trajectory for the better, and the one
874 statistically significant result we do find is more parsimoniously explained by
875 which economies self-select into SBFN membership than by what the policy
876 itself does once adopted. This is, we think, a genuinely different and com-
877plementary piece of evidence to Sections 5.2–5.7’s firm-level results, not a
878 restatement of them: different data source, different unit of analysis, differ-
879ent outcome constructs, and, for the first time in this paper, an identification
880 strategy that exploits exogenous variation in adoption *timing* rather than re-
881lying on cross-sectional comparison across adopters and non-adopters. That
882 it converges on the same substantive conclusion – no clean evidence of a ben-
883eficial institutional effect – from a design built specifically to meet a bar the
884 paper’s firm-level evidence cannot meet, strengthens rather than substitutes
885 for the paper’s central finding.

886 7. Discussion

887 Put in plain economic terms, the paper’s central finding is this: a firm
888 holding a bank credit line is associated with an adoption probability roughly
889 13 percentage points higher than an otherwise-similar firm without one—an

890 association net of observed controls, not a causally identified effect (Limita-
891 tion 1)—and that gap is essentially unchanged whether the firm sits in an
892 SBFN adopter or non-adopter, or across the full range of regulatory qual-
893 ity (Figure 4). Firms in SBFN-adopting economies are, on average, 15–20
894 percentage points *less* likely to have adopted a green practice at all—a level
895 gap, not a slope gap.

896 This convergence is informative precisely because of how differently the
897 evidence could have failed to agree. The primary and global samples share al-
898 most no economies, span different half-decades, use different outcome items,
899 and reach institutional settings—high-income, non-SBFN economies such as
900 Australia and Germany—the primary sample cannot touch. Section 6’s
901 country-level staggered event-study adds a third, independent test on top
902 of this: a different data source, a different unit of analysis, and the paper’s
903 only design that exploits exogenous variation in adoption *timing*. That it
904 too finds no clean evidence of a beneficial institutional effect—a precise null
905 on renewable-energy share, and a CO2 association that shrinks substantially
906 once development stage is controlled for—is the hardest-to-dismiss evidence
907 that this is a real feature of the data, not an artifact of any one sample, level
908 of analysis, or estimator. (This is a distinct claim from H2/H3’s firm-level
909 moderation null—a main effect at country level, not a slope change at firm
910 level—but evidence of the same kind.)

911 This complicates rather than confirms the implicit premise of a green-
912 credit-policy literature disproportionately built on China’s unusually mature,
913 well-resourced guidelines (Li et al., 2023; Huang et al., 2023; Dai et al., 2025).
914 Our sample spans every stage of SBFN’s own progression framework, from

915 Jordan and the Kyrgyz Republic’s initial “Commitment” to Mongolia’s “Ad-
916 vancing,” and across that range we find no evidence that formal adoption
917 changes how credit access translates into green investment. The causal for-
918 est’s feature-importance decomposition (Section 5.4) offers one redirection
919 rather than a tidy answer: the heterogeneity that does exist is overwhelm-
920 ingly a firm-size phenomenon (75.6% of explained variance) rather than in-
921 stitutional (regulatory quality 19.9%, SBFN 2.6%)—though it does not re-
922 solve into a clean size gradient, so we can say where the heterogeneity sits
923 without yet saying in which direction it runs. If the binding constraint is
924 which firms can absorb a green investment’s fixed costs once financed, rather
925 than whether a supervisory guideline nudges banks to lend more broadly, a
926 country-level mandate on the extensive margin of lending may be aimed at
927 the wrong margin regardless of its own credibility.

928 “No interaction detected” is not the same claim as “no institutional com-
929 plementarity exists,” and we do not want the former to be read as settling
930 the latter. Two alternative readings stay live: a linear interaction may be
931 the wrong functional form for a genuinely threshold-shaped complementarity,
932 given SBFN’s own graded progression framework and our binary coding of
933 it (Limitation 2); and the Worldwide Governance Indicators may simply not
934 proxy the banking-supervisory capacity a green-lending guideline depends
935 on (Limitation 5). We think our preferred reading—no complementarity
936 through this channel—is best supported by the evidence, but hold it as the
937 best current account rather than a claim that rules the alternatives out.

938 Two further limits are worth naming directly. First, SBFN guidelines
939 could in principle be working on the *extensive* margin of who obtains credit

940 at all, rather than the *intensive* margin our interaction terms test; testing
941 that would require modeling credit access itself as a function of SBFN status,
942 which our demand-side design does not do. Second, every outcome in this
943 paper is firm-reported, and we cannot rule out a compliance-reporting-versus-
944 practice gap between what firms report and what banks report to their own
945 supervisor—a limit no firm-level survey instrument can fully resolve on its
946 own. Read together with Section 6, whose null is, to our knowledge, the first
947 test of the SBFN framework’s own theory of change against outcomes on
948 the ground rather than its own tracked policy-stage progression, both point
949 the same way: a framework can look like it is working by every measure the
950 network itself tracks while the channel it is meant to move remains, on this
951 evidence, unmoved.

952 The one interaction that clears a conventional significance threshold—a
953 negative credit $\times$ SBFN effect for overdraft rather than term-credit access
954 (Section 5)—is more consistent with this redirection than with a simple policy
955 failure: overdrafts finance working capital, not the long-horizon expenditure
956 our outcome measures, so a supervisor successfully steering credit toward
957 term instruments would produce exactly this pattern. We flag it as suggestive
958 only: it is one significant coefficient among many tested, and does not survive
959 a Benjamini–Hochberg correction at any conventional false discovery rate
960 (Supplementary Table S3).

961 Read against the broader themes of state capacity and policy credibility,
962 our results sit closer to a state-capacity-is-necessary-but-firm-level-frictions-
963 bind-independently story than a straightforward complementarity one. Reg-
964 ulatory quality raises the baseline propensity to adopt green practices, but

965 is not the missing ingredient that lets a green banking guideline reshape
966 the credit-to-green channel specifically. The binding constraint looks less
967 like whether the supervising jurisdiction can make its guideline credible, and
968 more like whether firms below a certain scale can absorb a green investment's
969 fixed costs once credit is available—a question one level down from the one
970 the SBFN framework is designed to answer.

971 *7.1. Lessons for policy and for evaluation*

972 Three lessons follow for those who design and evaluate green finance
973 programmes. First, monitoring adoption is not monitoring effect. SBFN's
974 progress framework scores members on the guidelines they issue and the
975 stage they reach. On our evidence, being further along that framework did
976 not coincide with any change in the firm-level margin the guidelines target.
977 Programmes of this kind need outcome indicators alongside output indica-
978 tors: at a minimum, the share of bank credit financing green investment, and
979 firm-level adoption measured before and after a guideline takes effect.

980 Second, the data needed to evaluate firm-level effects properly do not yet
981 exist. The Enterprise Surveys' Green Economy module has been fielded
982 to most economies once. Re-fielding a comparable module in the same
983 economies would allow the within-country before-and-after comparison that
984 this paper could run only at country level, and would let future evaluations
985 locate where in the theory of change (Section 2.1) the chain breaks.

986 Third, if the binding constraint lies with firms rather than with banks
987 or supervisors, guidelines aimed at bank lending will need complements that
988 act on firms directly. The firm-size heterogeneity in Section 5.4 points in
989 this direction, though it is exploratory and does not resolve into a clean size

990 gradient.

991 8. Conclusion

992 This paper evaluated the wave of green banking policy adoption coor-
993 dinated through the Sustainable Banking and Finance Network against its
994 own theory of change: whether adoption strengthens the relationship between
995 firm-level access to finance and green investment, whether that depends on
996 the regulatory capacity available to make the policy credible, and whether
997 adopting economies’ aggregate environmental outcomes improve. Using har-
998 monized WBES data across 41 economies (2018–2020), a four-country waste-
999 minimization check, and an independently assembled 162-economy global
1000 sample (2021–2026) reaching for the first time into high-income, non-SBFN
1001 economies, we apply four escalating estimators and find five consistent re-
1002 sults. First, bank credit access is robustly associated with green practice
1003 adoption, surviving every specification, subsample, and alternative-outcome
1004 check, and replicating in both the small waste-minimization sample and the
1005 full global sample. Second, neither SBFN adoption nor regulatory quality
1006 moderates that relationship under any estimator, on either sample, though
1007 both shift the general level of adoption directly in the primary sample. Third,
1008 where effect heterogeneity does exist, it sits at the level of firm size rather
1009 than country institutions. Fourth, this pattern replicates on a sample sharing
1010 almost no economies with the primary one, under a materially sharper iden-
1011 tification strategy. Fifth, and most importantly for the paper’s credibility, an
1012 independent country-level staggered event-study (Section 6)—a main-effect
1013 test, not a repeat of the firm-level moderation null—likewise finds no clean

1014 evidence that SBFN adoption improves aggregate environmental outcomes,
1015 the single strongest evidence that this null is a real feature of the data rather
1016 than an artifact of any one sample, data source, level of analysis, or estimator.

1017 For policy, the implication is not that green banking guidelines are in-
1018 effective, but that our evidence does not support treating them as a lever
1019 for widening the green-investment gap between financed and unfinanced
1020 firms—the margin most of this literature implicitly assumes they operate
1021 on. If our tentative loan-tenor reading of the overdraft result is correct,
1022 policymakers may find more traction focusing on the term-lending instru-
1023 ments that actually finance green capital expenditure, and on complementary
1024 measures—technical assistance, matching grants, credit guarantees—that ad-
1025 dress smaller firms’ absorptive-capacity constraints directly.

1026 We close with seven limitations. First, both firm-level samples are sin-
1027 gle cross-sections (Section 4 explains why a genuine panel does not exist for
1028 this outcome): they cannot rule out reverse causality or unobserved coun-
1029 try characteristics correlated with both SBFN adoption and baseline green-
1030 adoption propensity, and the global sample’s fixed effects sharpen this only
1031 for the interaction terms, not the credit main effect. Second, SBFN mem-
1032 bership is a binary proxy for what the network’s own framework treats as
1033 a graded process; the global sample’s regional roster entries (the CEMAC
1034 and Eastern Caribbean groupings, Section 4) additionally rest on an infer-
1035 ence about coverage rather than an explicit per-country statement. Third,
1036 the four-economy waste-minimization check cannot re-test the SBFN con-
1037 trast (three of its four economies are members) and combines non-identical
1038 outcome items by construction of WBES’s own shortened module design—a

1039 data-availability constraint, not a methodological choice. Fourth, the global
1040 sample’s CO2-monitoring outcome is a single item, narrower than the pri-
1041 mary sample’s seven-item composite; that the same pattern holds on it is
1042 reassuring but does not establish it would hold on a richer measure. Fifth,
1043 the Worldwide Governance Indicators measure is a broad governance indica-
1044 tor, not a banking-supervision-specific capacity measure. Sixth, our design
1045 tests only the intensive margin of the credit-to-green channel and cannot
1046 rule out that SBFN policy operates on the extensive margin of who obtains
1047 credit in the first place—a margin our realized-access measure cannot ob-
1048 serve, since some firms decline formal credit altogether (Jin, 2026)—nor dis-
1049 tinguish firm-reported practice adoption from a bank-compliance-reporting
1050 gap (Section 7). Seventh, Section 6’s event-study is an ecological-level test
1051 of a related but distinct question; its one significant result (CO2 per capita)
1052 is netted against a single covariate, and we cannot rule out that a richer set
1053 of development-stage controls would shrink it further, or that some genuine,
1054 small, confounded relationship remains. We leave each of these to future
1055 work.

1056 **Statement on the use of AI tools**

1057 During the preparation of this work the author(s) used AI-assisted tools
1058 to help polish the writing. After using these tools, the author(s) reviewed
1059 and edited the content as needed and take full responsibility for the content
1060 of the published article.

1061 **Supplementary material**

1062 The full per-economy SBFN membership and regulatory-quality cod-
1063 ing for both samples, Bayesian convergence diagnostics, the global sam-
1064 ple’s SBFN membership by region, and the additional robustness checks and
1065 multiple-testing correction referenced above (Section 5.5, 5.6) are reported
1066 in the Supplementary Material accompanying this submission.

1067 *Data availability*

1068 Data in support of the findings of this study is available from the cor-
1069 responding author upon reasonable request, subject to the World Bank En-
1070 terprise Surveys’ own terms of use for the underlying firm-level microdata
1071 (enterprisesurveys.org, free registration required), including the standard-
1072 ized cross-country database and Productivity Database underlying the global
1073 sample. The country-level SBFN policy-status (including the SBFN data
1074 portal’s member roster underlying the global sample) and Worldwide Gover-
1075 nance Indicators datasets constructed for this paper, the country-year World
1076 Development Indicators panel and Callaway-Sant’Anna event-study code un-
1077 derlying Section 6 (all freely redistributable, no registration required), to-
1078 gether with the full analysis and replication code, are available in a public
1079 repository. The repository link is withheld from this anonymized manuscript
1080 to preserve double-anonymized review and will be supplied to the editors on
1081 request and published with the article.

1082 **References**

1083 Acemoglu, D., Johnson, S., & Robinson, J. A. (2001). The colonial origins of
1084 comparative development: An empirical investigation. *American Economic*

- 1085 *Review*, 91(5), 1369–1401. <https://doi.org/10.1257/aer.91.5.1369>
- 1086 Aristei, D., & Gallo, M. (2023). Green management, access to credit, and
1087 firms' vulnerability to the COVID-19 crisis. *Small Business Economics*,
1088 1–33. <https://doi.org/10.1007/s11187-023-00759-1>
- 1089 Asif, M., Dhamija, A., Khan, W., Johri, A., & Wasiq, M. (2025). Factors af-
1090 fecting adoption of green practices by food firms: Evidences from the World
1091 Bank Enterprises Survey. *Clean Technologies and Environmental Policy*,
1092 27(9), 4841–4858. <https://doi.org/10.1007/s10098-024-03122-4>
- 1093 Athey, S., & Imbens, G. W. (2019). Machine learning methods that
1094 economists should know about. *Annual Review of Economics*, 11(1), 685–
1095 725. <https://doi.org/10.1146/annurev-economics-080217-053433>
- 1096 Athey, S., Tibshirani, J., & Wager, S. (2019). Generalized random forests.
1097 *The Annals of Statistics*, 47(2), 1148–1178. [https://doi.org/10.1214/](https://doi.org/10.1214/18-AOS1709)
1098 18-AOS1709
- 1099 Bambe, B. W. W., Combes, J.-L., Kouassi, B.-V., & Schwartz, S. (2026).
1100 Environmental policy stringency and firm efficiency in developing coun-
1101 tries. *Small Business Economics*, 66(3), 1163–1182. [https://doi.org/](https://doi.org/10.1007/s11187-025-01128-w)
1102 10.1007/s11187-025-01128-w
- 1103 Beck, T., Demirgüç-Kunt, A., & Maksimovic, V. (2005). Financial and legal
1104 constraints to growth: Does firm size matter? *The Journal of Finance*,
1105 60(1), 137–177. <https://doi.org/10.1111/j.1540-6261.2005.00727.>
- 1106 **x**

- 1107 Beck, T., Demirgüç-Kunt, A., & Maksimovic, V. (2008). Financing pat-
 1108 terns around the world: Are small firms different? *Journal of Finan-*
 1109 *cial Economics*, 89(3), 467–487. [https://doi.org/10.1016/j.jfineco.](https://doi.org/10.1016/j.jfineco.2007.10.005)
 1110 2007.10.005
- 1111 Callaway, B., & Sant’Anna, P. H. C. (2021). Difference-in-differences with
 1112 multiple time periods. *Journal of Econometrics*, 225(2), 200–230. [https:](https://doi.org/10.1016/j.jeconom.2020.12.001)
 1113 [//doi.org/10.1016/j.jeconom.2020.12.001](https://doi.org/10.1016/j.jeconom.2020.12.001)
- 1114 Cameron, A. C., & Miller, D. L. (2015). A practitioner’s guide to cluster-
 1115 robust inference. *Journal of Human Resources*, 50(2), 317–372. [https:](https://doi.org/10.3368/jhr.50.2.317)
 1116 [//doi.org/10.3368/jhr.50.2.317](https://doi.org/10.3368/jhr.50.2.317)
- 1117 Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C.,
 1118 Newey, W., & Robins, J. (2018). Double/debiased machine learning for
 1119 treatment and structural parameters. *The Econometrics Journal*, 21(1),
 1120 C1–C68. <https://doi.org/10.1111/ectj.12097>
- 1121 Chi, Y., & Yang, Y. (2023). Green finance and green transition by enterprises:
 1122 An exploration of market-oriented governance mechanisms. *Borsa Istanbul*
 1123 *Review*, 23(3), 628–646. <https://doi.org/10.1016/j.bir.2023.01.003>
- 1124 Costa, H., Demmou, L., Franco, G., & Lamp, S. (2024). The role of financing
 1125 constraints and environmental policy on green investment. *Economics Let-*
 1126 *ters*, 239, Article 111741. [https://doi.org/10.1016/j.econlet.2024.](https://doi.org/10.1016/j.econlet.2024.111741)
 1127 111741
- 1128 Dai, Q., He, J., Guo, Z., Zheng, Y., & Zhang, Y. (2025). Green finance for
 1129 sustainable development: Analyzing the effects of green credit on high-

- 1130 polluting firms' environmental performance. *Humanities and Social Sci-*
1131 *ences Communications*, 12(1), Article 854. [https://doi.org/10.1057/](https://doi.org/10.1057/s41599-025-05218-8)
1132 [s41599-025-05218-8](https://doi.org/10.1057/s41599-025-05218-8)
- 1133 De Haas, R., Martin, R., Muûls, M., & Schweiger, H. (2025). Managerial
1134 and financial barriers to the green transition. *Management Science*, 71(4),
1135 2890–2921. <https://doi.org/10.1287/mnsc.2023.00772>
- 1136 Demirgüç-Kunt, A., & Maksimovic, V. (1998). Law, finance, and firm growth.
1137 *The Journal of Finance*, 53(6), 2107–2137. [https://doi.org/10.1111/](https://doi.org/10.1111/0022-1082.00084)
1138 [0022-1082.00084](https://doi.org/10.1111/0022-1082.00084)
- 1139 Gao, D., Mo, X., Duan, K., & Li, Y. (2022). Can green credit policy pro-
1140 mote firms' green innovation? Evidence from China. *Sustainability*, 14(7),
1141 Article 3911. <https://doi.org/10.3390/su14073911>
- 1142 Gelman, A., & Hill, J. (2007). *Data analysis using regression and multi-*
1143 *level/hierarchical models*. Cambridge University Press. [https://doi.org/](https://doi.org/10.1017/CB09780511790942)
1144 [10.1017/CB09780511790942](https://doi.org/10.1017/CB09780511790942)
- 1145 Gelman, A., Hill, J., & Vehtari, A. (2020). *Regression and other stories*.
1146 Cambridge University Press. <https://doi.org/10.1017/9781139161879>
- 1147 Ghosh, D., & Dutta, M. (2022). Environmental behaviour under credit
1148 constraints – evidence from panel of Indian manufacturing firms. *Struc-*
1149 *tural Change and Economic Dynamics*, 63, 490–500. [https://doi.org/](https://doi.org/10.1016/j.strueco.2022.07.004)
1150 [10.1016/j.strueco.2022.07.004](https://doi.org/10.1016/j.strueco.2022.07.004)

- 1151 Goodman-Bacon, A. (2021). Difference-in-differences with variation in treat-
 1152 ment timing. *Journal of Econometrics*, 225(2), 254–277. [https://doi.](https://doi.org/10.1016/j.jeconom.2021.03.014)
 1153 [org/10.1016/j.jeconom.2021.03.014](https://doi.org/10.1016/j.jeconom.2021.03.014)
- 1154 Hall, P. A., & Soskice, D. (2001). *Varieties of capitalism: The institutional*
 1155 *foundations of comparative advantage*. Oxford University Press.
- 1156 Huang, Z., Gao, N., & Jia, M. (2023). Green credit and its obstacles:
 1157 Evidence from China’s green credit guidelines. *Journal of Corporate*
 1158 *Finance*, 82, Article 102441. [https://doi.org/10.1016/j.jcorpfin.](https://doi.org/10.1016/j.jcorpfin.2023.102441)
 1159 [2023.102441](https://doi.org/10.1016/j.jcorpfin.2023.102441)
- 1160 International Finance Corporation / Sustainable Banking and Fi-
 1161 nance Network (2025). *Sustainable Banking and Finance Network*
 1162 *global progress report 2025*. World Bank Group / IFC. [https:](https://documents1.worldbank.org/curated/en/099634202132617625/pdf/IDU-3c06eadd-dd50-446d-a6d0-3fe3907bf5d5.pdf)
 1163 [//documents1.worldbank.org/curated/en/099634202132617625/](https://documents1.worldbank.org/curated/en/099634202132617625/pdf/IDU-3c06eadd-dd50-446d-a6d0-3fe3907bf5d5.pdf)
 1164 [pdf/IDU-3c06eadd-dd50-446d-a6d0-3fe3907bf5d5.pdf](https://documents1.worldbank.org/curated/en/099634202132617625/pdf/IDU-3c06eadd-dd50-446d-a6d0-3fe3907bf5d5.pdf)
- 1165 Jia, S., Farooq, U., Alomair, A., & Aljughaiman, A. A. (2026). Credit
 1166 with a climate purpose: Does green credit policy accelerate sustainable
 1167 production in BRICS? *Borsa Istanbul Review*, 26(3), Article 100817.
 1168 <https://doi.org/10.1016/j.bir.2026.100817>
- 1169 Jin, W. (2026). Choosing not to borrow: Imprinting effects of informality on
 1170 firms’ credit self-rationing. *Borsa Istanbul Review*, 26(2), Article 100785.
 1171 <https://doi.org/10.1016/j.bir.2026.100785>
- 1172 Kalantzis, F., Schweiger, H., & Dominguez, S. (2022). *Green investment*

1173 *by firms: Finance or climate driven?* EBRD Working Paper No. 268.
1174 <https://doi.org/10.2139/ssrn.4132545>

1175 Khairunnessa, F., Vazquez-Brust, D. A., & Yakovleva, N. (2021). A review
1176 of the recent developments of green banking in Bangladesh. *Sustainability*,
1177 13(4), Article 1904. <https://doi.org/10.3390/su13041904>

1178 La Porta, R., Lopez-de-Silanes, F., Shleifer, A., & Vishny, R. (1998). Law
1179 and finance. *Journal of Political Economy*, 106(6), 1113–1155. <https://doi.org/10.1086/250042>

1181 Levine, R. (2005). Finance and growth: Theory and evidence. In P. Aghion,
1182 & S. Durlauf (Eds.), *Handbook of economic growth* (Vol. 1A, pp. 865–934).
1183 Elsevier. [https://doi.org/10.1016/S1574-0684\(05\)01012-9](https://doi.org/10.1016/S1574-0684(05)01012-9)

1184 Li, C., Feng, X., Li, X., & Zhou, Y. (2023). Effect of green credit policy on en-
1185 ergy firms' growth: Evidence from China. *Economic Research-Ekonomska*
1186 *Istraživanja*, 36(2). <https://doi.org/10.1080/1331677X.2023.2177701>

1187 Li, C., Liu, Z., Song, R., & Zhang, Y.-J. (2024). The impact of green credit
1188 guidelines on environmental performance: Firm-level evidence from China.
1189 *Technological Forecasting and Social Change*, 205, Article 123524. <https://doi.org/10.1016/j.techfore.2024.123524>

1191 Nguyen, M. P., & Phan, A. (2025). Impact of green credit policy on
1192 the operation of Vietnamese commercial banks: An empirical study us-
1193 ing difference-in-differences model. *PLOS One*, 20(10), Article e0333759.
1194 <https://doi.org/10.1371/journal.pone.0333759>

- 1195 Phan, Q. T. (2026). The impact of green economy practices on firm
1196 innovation using data from the World Bank Enterprise Surveys. *Dis-*
1197 *cover Sustainability*, 7(1), Article 848. [https://doi.org/10.1007/](https://doi.org/10.1007/s43621-026-03226-5)
1198 [s43621-026-03226-5](https://doi.org/10.1007/s43621-026-03226-5)
- 1199 Salvatier, J., Wiecki, T. V., & Fonnesbeck, C. (2016). Probabilistic program-
1200 ming in Python using PyMC3. *PeerJ Computer Science*, 2, Article e55.
1201 <https://doi.org/10.7717/peerj-cs.55>
- 1202 Shi, Y., Zhu, Q., & Khan, M. A. (2024). The efficacy of green finance for en-
1203 vironmental sustainability: Does control of corruption makes a difference?
1204 *Borsa Istanbul Review*, 24(6), 1179–1189. [https://doi.org/10.1016/j.](https://doi.org/10.1016/j.bir.2024.07.002)
1205 [bir.2024.07.002](https://doi.org/10.1016/j.bir.2024.07.002)
- 1206 Siewers, S., Martínez-Zarzoso, I., & Baghdadi, L. (2024). Global value chains
1207 and firms' environmental performance. *World Development*, 173, Article
1208 106395. <https://doi.org/10.1016/j.worlddev.2023.106395>
- 1209 Tran, T. T. C., Nguyen, L., & Do, T. K. (2026). The impact of fin-
1210 tech credit on green innovation: International evidence. *Journal of Eco-*
1211 *nomics and Development*, 28(2), 111–127. [https://doi.org/10.1108/](https://doi.org/10.1108/JED-05-2025-0177)
1212 [JED-05-2025-0177](https://doi.org/10.1108/JED-05-2025-0177)
- 1213 Truong, D. D. (2026). Does access to finance condition firms' green in-
1214 vestment responses to environmental pressure? Evidence from Viet-
1215 nam. *PLOS One*, 21(5), Article e0341960. [https://doi.org/10.1371/](https://doi.org/10.1371/journal.pone.0341960)
1216 [journal.pone.0341960](https://doi.org/10.1371/journal.pone.0341960)

- 1217 Ullah, B. (2025). Corporate environmental responsibility and financial con-
1218 straints for unlisted SMEs. *Global Finance Journal*, 67, Article 101176.
1219 <https://doi.org/10.1016/j.gfj.2025.101176>
- 1220 Ullah, W., Zubir, A. S. M., & Ariff, A. M. (2024). Exploring the mod-
1221 erating effect of regulatory quality on the relationship between finan-
1222 cial development and economic growth/economic volatility for developed
1223 and developing countries. *Borsa Istanbul Review*, 24(5), 934–944. [https:](https://doi.org/10.1016/j.bir.2024.04.015)
1224 [//doi.org/10.1016/j.bir.2024.04.015](https://doi.org/10.1016/j.bir.2024.04.015)
- 1225 Wager, S., & Athey, S. (2018). Estimation and inference of heterogeneous
1226 treatment effects using random forests. *Journal of the American Sta-*
1227 *tistical Association*, 113(523), 1228–1242. [https://doi.org/10.1080/](https://doi.org/10.1080/01621459.2017.1319839)
1228 [01621459.2017.1319839](https://doi.org/10.1080/01621459.2017.1319839)